

MASTERS OF MECHANICAL ENGINEERING / MECHATRONICS

VIRTUAL PRODUCT DEVELOPMENT SS-25

REPORT OF FINGER PRINTER

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Project Overview

The **Virtual Product Development (VPD)** course is a core part of the Master's program in Mechanical Engineering / Mechatronics at the University of Applied Sciences Kaiserslautern. Under the guidance of Professors Thomas Kilb and Matthias R. Leiner, the course emphasizes simulation-based product development, utilizing leading software such as Siemens NX, Altair Inspire, and Altair MotionView.

Throughout the semester, students undertake a hands-on project designed to apply theoretical knowledge and enhance their expertise with advanced simulation tools. These collaborative projects are completed in small, diverse groups of international students, fostering teamwork and practical skill development.

Group B consists of the following members: 1. Mithin Kumar Chithirala

2. Madana Gopala Reddy Konda

3. Yuvraj Ramchandra Labhade

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Phase 1 – Initial Development

This academic project is divided into three structured phases, each ending with a technical report and a formal presentation. The phased approach supports a logical progression from concept to realization, fostering interdisciplinary problem-solving within a Virtual Product Development (VPD) environment.

The central goal of Phase 1 is to conceptualize and initiate the design of a "finger printer", a custom end-effector intended for integration with the Tronxy X5SA Pro 3D printer. The finger printer will expand the capabilities of standard FDM printing by enabling interactions with flexible or repositionable materials and geometries—particularly those involving undercuts and complex overhanging structures.

As outlined by Kilb and Leiner (2025), the initial phase focuses on the early-stage development and technical definition of the finger printer system. The key deliverables include:

- Analysis of the FDM 3D printing process
- Gain basic knowledge about FDM 3D printing processes with undercuts
- Development goals (to ensure customer benefits for overhangs, undercuts, etc.)
- Actuator concept (including selection of suitable joints and actuators)
- Definition of functional and design space requirements
- Packaging concept
- CAD design (draft)
- Material selection
- Make or buy decisions (of standard parts)

Comprehensive Market and Technology Research

3D printing, particularly FDM, has rapidly grown in consumer and industrial use. Comprehensive research on printers, processes, materials, and finger mechanisms is essential.

This includes exploring actuator concepts and material choices. Understanding these areas enables effective design development. The goal is to combine additive manufacturing with mechanical design to build a strong foundation for innovation.

Overview of Additive Manufacturing and FDM Technology

Additive manufacturing (AM), commonly known as 3D printing, is a transformative technology that builds physical objects layer by layer from digital 3D model data, enabling the creation of complex geometries that are difficult or impossible to produce using traditional methods (Gibson, Rosen, and Stucker 2015).

The process starts with a detailed CAD design, which is then sliced into thin layers by specialized software to guide the precise deposition of materials such as thermoplastics, metals, or composites. Among various AM techniques, Fused Deposition Modelling (FDM) is the most widely adopted due to its affordability, ease of use, and versatility across applications (Chua, Leong, and Lim 2010).

This project aims to design a product compatible with existing FDM print heads and material systems, ensuring effective integration and reliable printing results.

Fundamentals and Key Components of 3D Printing Technology

3D printing, also known as additive manufacturing, is a process that creates physical objects by building successive layers of material based on digital 3D models. The workflow begins with designing a CAD model, which is then converted into instructions through slicing software. The printer deposits materials such as plastics, metals, or ceramics layer by layer, producing complex geometries with high precision (Coursera 2024).

This technology finds applications across numerous fields including healthcare (e.g., custom prosthetics and implants), aerospace (lightweight structural components), automotive (rapid prototyping and tooling), and consumer goods (customized products). The versatility and growing accessibility of 3D printing have transformed traditional manufacturing methods and opened new avenues in product innovation (Coursera 2024).

Essential Components of a 3D Printer:

- **Filament:** The raw thermoplastic or composite material supplied as a continuous coil, fed into the extruder for melting and deposition (Wong & Hernandez 2012).
- **Extruder:** Regulates the flow rate of the filament into the hot end to ensure consistent material deposition.
- **Motors:** Control precise movements along the X, Y, and Z axes and filament feeding mechanisms.
- **Nozzle:** The tip through which molten material is extruded onto the print bed.
- **Print Bed:** The surface where printing occurs, which can be heated to improve adhesion and print quality.
- **Cooling Fans:** Manage thermal regulation to prevent premature melting and help solidify layers rapidly.
- **Hot End:** The heated component that melts filament before extrusion, often comprising a heat break, heat sink, and nozzle assembly.

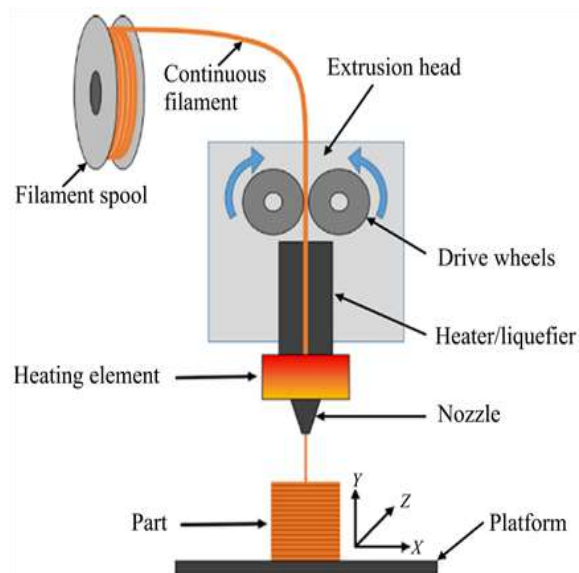


FIGURE 1:3D PRINTER PROCESS.

<https://www.scirp.org/journal/paperinformation?paperid=123592>



FIGURE 2:3D PRINTER.

Slicing and Print Path Generation in 3D Printing

Slicing is a critical pre-processing step in the 3D printing workflow that translates a digital 3D model into a machine-readable set of instructions. The slicer software divides the model into horizontal layers and generates a tool path that determines how each layer will be deposited, optimizing parameters such as print time, strength, and material usage (Gebhardt & Hötter 2021).

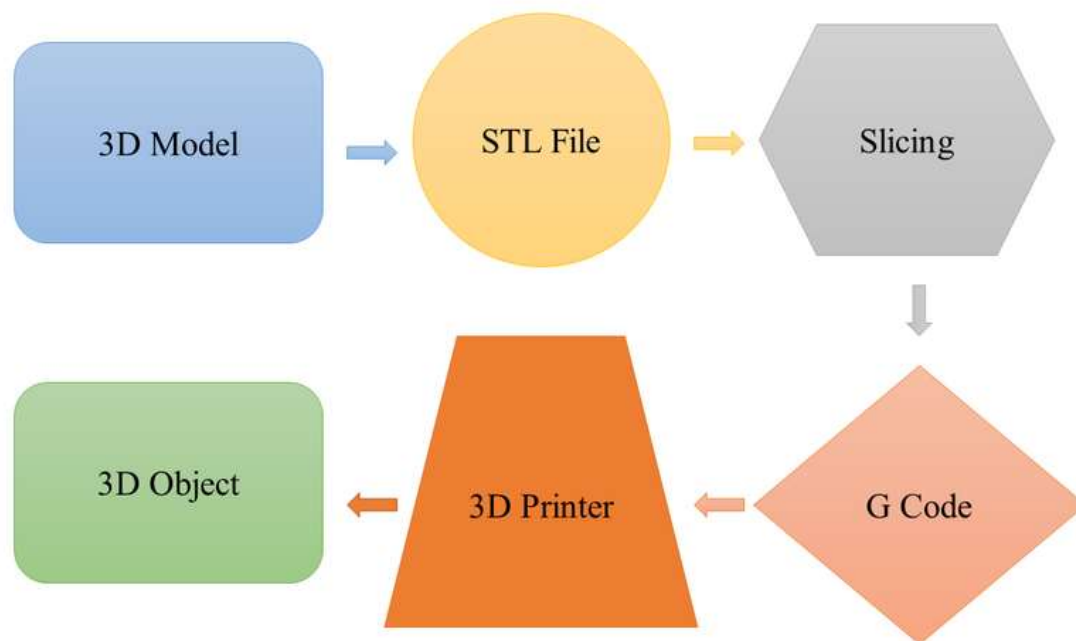


FIGURE 3: PROCESS OF SLICING.

<https://www.researchgate.net/figure>

Most slicers work with STL files, which define the surface geometry of the model using triangular facets. The output of the slicer is typically a G-code file, which includes instructions for printer movements, layer height, extrusion speed, and temperature settings. Advanced slicers also allow control over infill density, support structures, and print orientation, contributing significantly to the quality and mechanical performance of the printed part.

Additive Manufacturing

Fused Deposition Modelling (FDM) is one of the most widely used additive manufacturing techniques, particularly valued for its simplicity and cost-effectiveness. In this process, a continuous filament of thermoplastic material is heated to a semi-liquid state and extruded through a nozzle. The material is deposited layer by layer, either directly onto the build platform or atop previously printed layers.

The success of the FDM process relies heavily on the thermoplastic behaviour of the filament, which allows the layers to fuse during deposition and solidify upon cooling. Key factors such

as filament orientation, layer thickness, and width significantly influence the final mechanical properties of the printed part, including strength, adhesion, and surface finish.

Contemporary 3D Printing Materials

1. PLA – Polylactic Acid (Biopolymer Composites)

Polylactic acid (PLA) remains the most widely used thermoplastic in FDM, prized for its biodegradability, ease of processing, and compatibility with utility and biomedical applications. Recent studies enhance PLA by blending it with natural fibres such as bamboo, hemp, and cellulose to improve mechanical strength, stiffness, and sustainability. For example, PLA reinforced with microcrystalline cellulose achieved a threefold increase in tensile strength (~13.68 MPa) compared to neat. These PLA-bio composites are being explored for eco-friendly packaging, medical devices, and structural applications. (researchgate.net)

2. TPU & Other Flexible Elastomers

Thermoplastic elastomers (TPEs), especially TPU and TPC, offer flexibility, abrasion resistance, and resilience—qualities essential for soft robotics, wearable devices, and functional seals. A comparison of flexible filaments shows TPU excels at elasticity and printing reliability, while TPC is noted for its chemical and temperature resistance and PP for its lightweight durability (formfutura.com). TPU is extensively used in soft actuators, tubing, and medical applications due to its combined softness and mechanical strength.

3. Polycarbonate (PC) – High-Performance Thermoplastic

Polycarbonate stands out for its optical clarity, exceptional impact resistance (~70 MPa), and thermal stability (service temperatures up to ~110 °C). It's used in advanced applications like headlight prototypes, transparent housings, and engineering parts. Printing PC demands high temperatures (270–310 °C), an enclosed heated build chamber, and rigorous moisture control. (sovol3d.com)

4. HIPS – High Impact Polystyrene (Support Material)

High-impact polystyrene (HIPS) combines toughness and flexibility, making it ideal for use as a dissolvable support material in dual-extrusion setups. It dissolves cleanly in limonene, enabling smooth removal and high-quality surface finishes. (en.wikipedia.org)

5. ABS (Acrylonitrile Butadiene Styrene)

ABS is a petroleum-based thermoplastic known for its toughness and heat resistance (glass transition ~105 °C). It's commonly used for functional prototypes, automotive parts, and enclosures. However, printing ABS demands a heated bed and enclosure due to warping, and it emits potentially harmful fumes, requiring proper ventilation.

6. PETG (Polyethylene Terephthalate Glycol-modified)

PETG combines the user-friendliness of PLA with the strength and flexibility of ABS. It prints well at 220–250 °C with minimal warping, excellent layer adhesion, and chemical resistance. PETG is increasingly used for mechanical parts and outdoor applications, striking a balance between durability and ease. (sovol3d.com)

TABLE 1: TECHNICAL SPECIFICATION OF MATERIALS

Material	Print Speed (mm/s)	Density (g/cm ³)	Typical Printing Temp (°C)	Melting Temp (°C)	Notes
PLA	40 - 60	1.24 - 1.30	190 - 220	~150 - 160	Easy to print, low warping, biodegradable
TPU	15 - 30	1.10 - 1.21	210 - 240	~220 (softens)	Flexible, elastic, slower print speeds
Polycarbonate (PC)	30 - 60	1.20 - 1.22	260 - 310	~260 - 270	High strength, heat resistant, tricky print
HIPS	40 - 60	1.04 - 1.08	220 - 250	~220 - 240	Often used as support, dissolvable in limonene
ABS	40 - 60	1.04 - 1.07	220 - 250	~210 - 240	Strong, flexible, requires heated bed
PETG	40 - 70	1.27 - 1.38	230 - 250	~230 - 260	Tough, chemically resistant, easy to print

Direct vs Bowden Extrusion: Mechanisms in 3D Printing

3D printers mainly use two types of filament extrusion mechanisms: Direct Drive and Bowden Drive systems. Each has distinct mechanical layouts and advantages based on material type and desired print performance.

Direct Drive Extrusion System

In a Direct Drive system, the filament drive motor is mounted directly onto the print head, very close to the hot end. This compact design ensures minimal distance between the extruder and nozzle, enabling precise control over filament movement.

Advantages:

- Ideal for flexible filaments like TPU or TPE, as the shorter filament path prevents buckling.
- Enhanced retraction control, which helps reduce stringing and improves surface quality.
- Easier to tune for complex materials and more responsive extrusion behaviour.

Disadvantages:

- The motor and extruder components add weight to the print head, which can limit printing speed and cause more vibration during fast movements.
- May require more frequent calibration to avoid quality drops due to mechanical load.

Typical Use: Common in high-precision and flexible-material-focused printers. (Prusa Research).

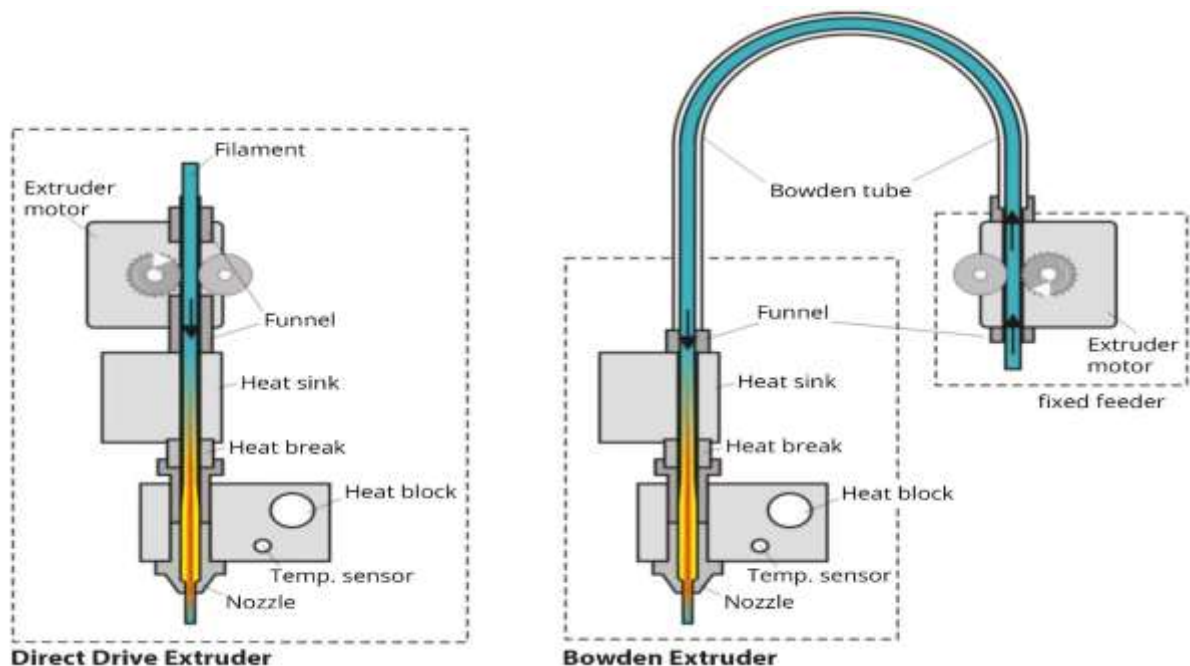


FIGURE 4:DIFFERENCE BETWEEN DIRECT AND BOWDEN EXTRUDER.

<https://zengo.eu/en/blog/fdm-fused-deposition-modeling>

Bowden Drive Extrusion System

In a Bowden setup, the extruder motor is mounted on the printer's frame, and the filament is guided to the hot end through a PTFE or Teflon tube. This reduces the weight of the print head and improves motion speed.

Advantages:

- Lighter print head allows for faster printing and more stable high-speed motion.
- More suitable for larger format printers where long filament paths are inevitable.

Disadvantages:

- Not well-suited for flexible filaments, as they tend to compress or buckle within the tube.
- Retraction settings are harder to calibrate, potentially leading to oozing or stringing.

- Slightly delayed filament response due to tubing elasticity.

Typical Use: Popular in high-speed printers like Creality Ender-3, where speed and simplicity are priorities. (Prusa Research).

TABLE 2: COMPARATIVE ANALYSIS OF 3D PRINTING TECHNOLOGIES

Technology	Build Volume (mm ³)	Materials	Print Speed	Extrusion System	Build Platform	Cooling System	Layer Resolution (μm)	Typical Applications
FDM (Fused Deposition Modelling)	Small (~150×150×150)	PLA, ABS, PETG	Moderate	Single extruder	Heated bed	Passive cooling	100–300	Prototyping, hobbyist models
SLA (Stereo lithography)	Medium (~300×300×300)	Photopolymer resin, ABS	Fast	Dual extruder	Non-heated bed	Active (fans)	25–100	Dental, jewellery, detailed parts
SLS (Selective Laser Sintering)	Large (~500×500×500)	Nylon, PETG	Very fast	Multi extruder	Flexible bed	Liquid cooling	50–150	Functional prototypes, aerospace
DLP (Digital Light Processing)	Customizable	Resin, Nylon	Adjustable	Direct extrusion	Auto-levelling bed	Advanced thermal management	25–75	Miniatures, dental, engineering
MJF (Multi Jet Fusion)	Variable	Polyamide resin	Adjustable	Bowden system	Powder bed	Self-regulated (powder insulation)	80–120	Industrial parts, automotive

The Role of Undercuts and Overhangs in 3D Printing

To fully grasp the challenges addressed in this project, it is essential to clearly define what undercuts are and why they represent a critical obstacle.

Undercuts are specific regions of a part that can only be accessed from below during the printing process. In practical terms, this means that the printer's nozzle must approach these areas from underneath the feature it is printing. While it might seem possible to reach these points from the same height as the target, doing so significantly increases the risk of the nozzle colliding with or scratching previously printed sections.

The main difficulty with printing undercuts arises from the fundamental working principle of most standard 3D printers. These printers build parts layer by layer from the bottom up, depositing material in a vertical, upward direction. To print an undercut, the nozzle would have to move top-down or from underneath, which is the exact opposite of the conventional printing

approach. As a result, standard 3D printers cannot create undercuts without support or specialized hardware.

Differentiating Undercuts from Overhangs

It is equally important to distinguish undercuts from overhangs, as they are often confused but represent very different printing challenges.

An overhang refers to a section of the part that extends horizontally outward, where there is no material directly underneath. However, unlike an undercut, an overhang remains connected to the main part. Standard 3D printers can usually handle overhangs, but often require special slicing software to adjust printing parameters or the addition of support structures below the overhanging areas. These supports prevent the material from sagging or collapsing during printing and are removed once the print is complete.

The project discussed here—the finger printer concept—aims to revolutionize this by enabling the printing of overhangs without needing any support structures. This would significantly reduce material waste and post-processing efforts.

Enhanced 3D Printing Through Multi-Axis Motion and Variable Printing Directions

Traditional Fused Deposition Modelling (FDM) 3D printers operate along three linear axes (X, Y, and Z), which inherently restricts the fabrication process to a single, fixed build direction. This limitation poses significant challenges such as the inability to print steep overhangs (typically beyond 45°–60°) without support structures and the impossibility of producing undercuts—features only accessible from beneath the build surface (Wang et al 2018).

These constraints result in compromised mechanical strength, suboptimal surface finish, and increased material usage for supports (Bächer et al 2020).



FIGURE 5: BÄCHER ET AL 2020

The introduction of multi-axis 3D printing technologies expands the design space, enabling the production of complex geometries that are unattainable with standard three-axis systems. By incorporating rotational axes to the printer's motion, the print head can approach the build from multiple angles, thereby reducing or eliminating the need for support structures and improving surface quality (Sun et al 2019). This advancement necessitates significant adaptations in both machine kinematics and print head design.



FIGURE 6: WÜTHRICH ET AL. 2021

This project aims to explore the capabilities and requirements of a multi-axis 3D printer, focusing on modifying an existing Tronxy print head by adding two rotational degrees of freedom. This modification is intended to facilitate printing at inclined angles between 20° and 70°, enhancing the printer's ability to produce overhangs and undercuts.

Multi-Axis 3D Printers: New Possibilities for Complex Printing

Most 3D printers available today use the basic three-axis system (X, Y, and Z). These work well for many applications, but they have major limitations. For example, standard printers can't handle steep overhangs or undercuts without using support material. This not only increases material waste but also affects surface quality and mechanical strength (Yang et al., 2020).

To solve these problems, researchers and hobbyists have started working on multi-axis 3D printers. These printers add extra rotating axes, which allow the print head or the print bed to tilt or rotate. This makes it easier to print parts from different angles and reduce or even eliminate supports (Yao et al., 2023). Although these systems are not yet widely available commercially, several promising prototypes have been developed.

1. Belt Printers (Commercial)

- **Nozzle tilt:** Always fixed at about 45°, enabling superior handling of overhangs compared to conventional printers.
- **Belt actuation:** Replaces the Z-axis with a moving conveyor belt, enabling continuous and extended part printing—useful for multiple or long parts.

- **Relevance to VPD project:** Excluded due to the conveyor mechanism, but the inclined nozzle concept remains a valuable reference.

2. RotBot (ZHAW, 2021)

- **Configuration:** Standard Prusa base with an added 4th rotational axis around the Z-axis.
- **Nozzle:** Tilted (~45°) to enhance overhang support—effectively allowing up to 90° overhang.
- **Mechanics:** Well-integrated motor and cabling on the rotating axis.
- **Strengths:** Improved angled-surface quality and proof-of-concept for single-axis rotation.
- **Limitations:** Does not allow undercut printing and lacks full multi-axis movement.

3. Dual-Revolute Axis Prototype (Andersons, 2023)

- **Design:** Two rotating joints—vertical and horizontal—driving the nozzle in versatile orientations.
- **Build:** Fully 3D-printed, with gear-driven rotation.
- **Focus:** Emphasis on UI/software rather than mechanical refinement.
- **Capabilities:** Supports angled overhangs, but cannot handle undercuts due to limited axis freedom.

4. HCC 3D Spherical Nozzle System (K, 2024)

- **Axes:** Vertical rotation at the XY base and secondary rotation following a spherical path.
- **Movement style:** Gimbal-like, allowing variable nozzle angles for better surface detail.
- **Strengths:** Flexible and good surface finish.
- **Drawbacks:** Heavy construction and incapability with undercuts.

5. PAX DH 135 – 5-Axis System (Mueller, ZHAW, 2021)

- **Axes:** Standard XYZ plus two rotational axes allowing 135° nozzle articulation.
- **Drive mechanism:** Direct-drive joints for precise control.
- **Advantages:** High surface quality, structural stability.
- **Disadvantages:** Bulky design, still unable to tackle undercuts, and limited compactness despite 3D-printed components.

Gimbal-Like Nozzle Movement

A **gimbal mechanism** enables rotation about two orthogonal axes—typically pitch and yaw. Applying this to a 3D printer nozzle:

- **Pitch axis:** Tilts the nozzle up/down.
- **Yaw axis:** Rotates nozzle left/right.

- **Benefit:** Offers a clear mechanical analogy for how adding two rotational axes can help the nozzle reach overhangs and potentially undercuts.

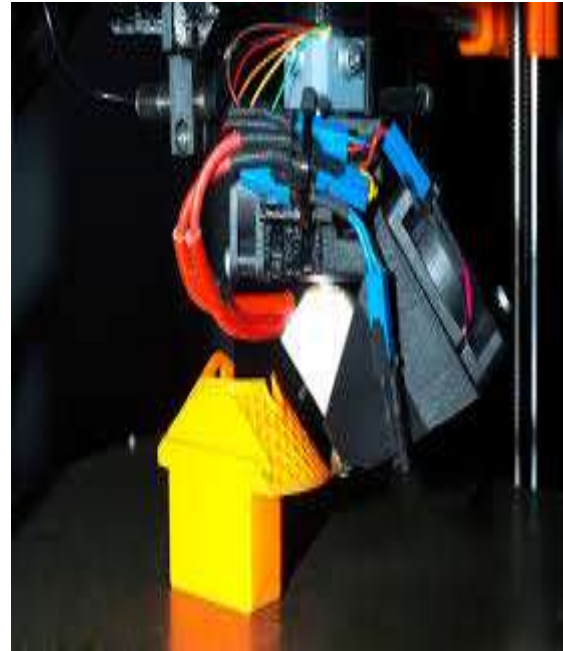
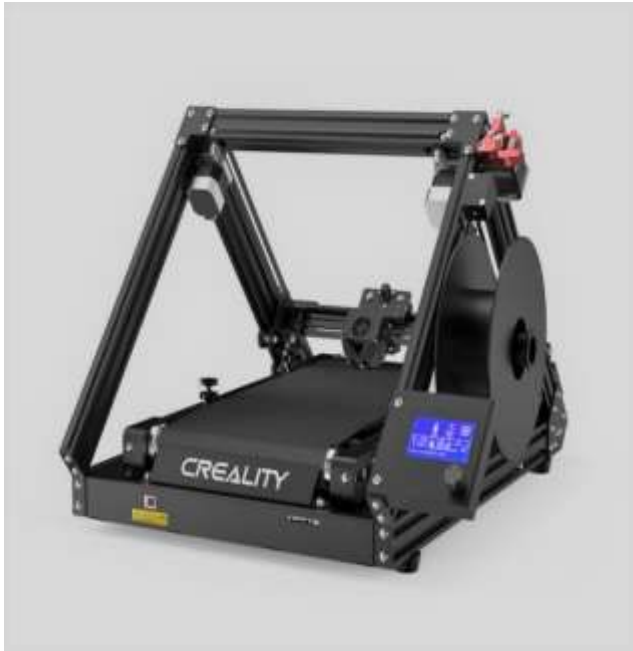


FIGURE 7: FIG 7 : ABOUT AXIS.

3dprintingindustry.com

www.am-expo.ch



FIGURE 8: NOZZLE MOMENT.

www.kuka.com

www.instructables.com

Design Inspirations from Human Fingers

To develop a mechanical finger, a good starting point is understanding the structure of a human finger. Each finger contains one ball joint and two hinge joints, allowing flexible movement. Interestingly, the muscles that control finger motion are located in the hand or forearm rather than directly at the joints.

Prosthetic fingers have been developed for many years, and these devices vary widely in terms of price, materials, types of actuators, and overall weight. Since prosthetic fingers face similar challenges—such as needing to be lightweight, strong, and capable of wide movement—they provide valuable insights for our VPD project.

From a manufacturing perspective, most prosthetic fingers use 3D printing, with only a small number made from steel. 3D printing offers benefits like reduced costs and the ability to produce complex shapes easily. Therefore, our project will focus on 3D printing finger components using common, inexpensive materials like PLA or ABS.



FIGURE 9: FINGER

. janelangille.com



FIGURE 10:3D ARM

www.myminifactory.com

Motion and Actuation Concepts for Fingerprint Head Design

Finger Movement and Joint Types

To design flexible and functional components for a 3D printer, robotic fingers offer a helpful reference. These systems are typically more adaptable than rigid mechanical designs. Robotic fingers use different types of joints—mainly **revolute (rotational)** or **prismatic (sliding)** joints—and sometimes a combination of both. In our project, we are applying lessons learned from robotic kinematic systems to develop a motion-capable fingerprinting head.

Kinematics is the study of motion without focusing on forces. In this work, we focus on the geometry and movement of the parts involved, especially the relation between the end-effector (the moving part, like a finger or print head) and the base structure.

Kinematics is divided into two types:

- **Forward Kinematics (FK):** Calculates the position and orientation of the end-effector from known joint positions.
- **Inverse Kinematics (IK):** Determines the joint positions needed to reach a specific end-effector position. IK is more complex and often needs extra constraints to find accurate solutions.

Actuation Systems in Finger-like Mechanisms

To move robotic parts, different types of actuators are used. An actuator converts energy into mechanical motion. In our context, they are used to move the joints of the fingerprint head in a controlled way.

Below are three common types of actuators:

1. Electric Actuators

Electric actuators use electric motors to produce motion. They are the most commonly used in 3D printers due to their precision, speed, and compact size. Though they don't provide the highest force, their ease of control and affordability make them a popular choice.

Example Application: Stepper motors in standard 3D printers for X, Y, and Z motion.

2. Pneumatic Actuators

These actuators use compressed air to produce movement. They are lightweight and fast, making them suitable for automation tasks. However, they require air supply systems and offer less power compared to hydraulic systems.

Example Application: Pick-and-place robots in factories that use air pressure for quick movements.

3. Hydraulic Actuators

Hydraulic actuators use pressurized fluid for high-force motion. They are powerful and durable, typically found in heavy-duty machinery. However, they are bulky, more expensive, and can be messy due to potential fluid leaks.

Example Application: Industrial robot arms used in construction or mining.

Types of Joints Based on Kinematic Design

Joints in mechanical systems can be classified according to their movement, defined by their kinematic design. Each joint type has one or more degrees of freedom (DoF), which describe how the joint can move:

1. Linear Joints: Allow straight-line movement along one axis.
2. Revolute Joints: Enable rotation around a fixed axis.
3. Spherical Joints: Provide multi-directional rotational movement, like a ball-and-socket.

For our 3D printer fingerprint head, the electric motor-driven revolute joint is the most suitable because it offers precise angular control and simplicity in design.

Choosing a Motor?

When selecting a motor for a robotic or automated application, engineers typically consider three main factors:

- Speed Range: Minimum and maximum speeds, including how quickly the motor can accelerate.
- Torque Characteristics: Maximum torque and how torque changes with speed.
- Precision and Repeatability: How accurately the motor can position without relying on sensors or feedback loops.

Other considerations include size, weight, cost, and power consumption. The most common motor types for small to medium actuators are brushed DC motors, brushless DC (BLDC) motors, and stepper motors. In some cases, pneumatic or hydraulic actuators may be preferable depending on the application. (Robotics Actuators and Motors Overview, Kumar et al 2022)

Stepper Motors

How They Work:

Stepper motors rotate in fixed increments or steps. Each electrical pulse moves the motor shaft by a specific angle, making them easy to control with open-loop systems (no sensor feedback needed).

Typical Uses:

- 3D printers for precise layer movement
- CNC machines for exact positioning
- Basic robotic arms for repetitive tasks
- Automated assembly lines requiring repeatability

Benefits:

- Simple control without complex feedback systems
- Cost-effective compared to servo motors
- High holding torque when stationary, which helps maintain position

Drawbacks:

- Consume power constantly, even when not moving
- Limited speed and torque at higher rotational speeds
- Can overheat under heavy loads
- May lose steps under excessive load, causing position errors

(Motor Control Technologies, Lee and Park, 2023)

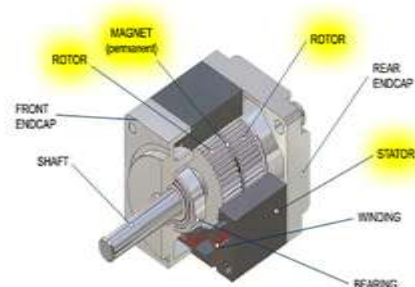


FIGURE 11: STEPPER MOTOR

www.amci.com

Servo Motors

How They Work:

Servo motors operate with closed-loop control, using sensors (encoders or potentiometers) to continuously monitor and adjust the shaft position for precise movement.

Typical Uses:

- Advanced robotics requiring fine control
- Conveyor systems with variable speed and positioning needs
- Aerospace applications demanding reliability and accuracy
- CNC machining with high-speed precision

Benefits:

- High accuracy and smooth motion due to feedback control
- Strong torque output across a wide speed range
- More efficient power usage by adjusting power as needed
- Fast dynamic response to control inputs

Drawbacks:

- More complex and costly control systems
- Requires additional hardware like encoders and controllers
- Maintenance and tuning can be more demanding

(Precision Actuation Systems, Zhang et al 2021)



FIGURE 12: MOTOR COMPONENTS

www.amci.com



FIGURE 13: SERVO MOTOR

Joint Types Used

In this study, two revolute joints are employed within the mechanical system due to their straightforward design, reliability, and efficiency in facilitating rotational motion within a compact space.

This approach is particularly advantageous for applications requiring precise rotational control with minimal mechanical losses, especially when high gear reduction ratios are not essential.

Concept Development

The initial stage of developing the concept revolves around identifying what makes the product stand out in the marketplace and understanding the needs that are yet to be met within the 3D-printing industry. A thorough analysis of existing market insights and project objectives helps clarify these aspects.

Key factors were identified through collaborative discussions, focusing on the core elements that will shape the design:

- **Stability:** Achieving high-quality printed parts, especially for finger components, depends on effective layer bonding. Optimizing the nozzle angle during printing enhances the fusion of layers, which is critical for complex geometries.
- **Support-Free Printing:** Removing the need for support structures can improve efficiency by reducing material consumption and printing time. It also minimizes post-processing efforts and waste production. However, achieving support-free prints often requires specialized slicing software beyond what conventional printers provide.
- **Surface Quality:** The finish of printed parts is a major selling point for both printers and filament manufacturers. Since printers often work with various filaments, ensuring the nozzle maintains a perpendicular orientation to the part surface is crucial to maximize surface quality.
- **Reachability:** Preserving the print volume of the existing Tronxy 3D printer is a priority. Any necessary workspace reductions should be minimal and preferably along the Z-axis, as it offers the greatest print height. The aim is to maintain at least 250 mm along the X and Y axes and 300 mm along the Z axis, with larger dimensions preferred.

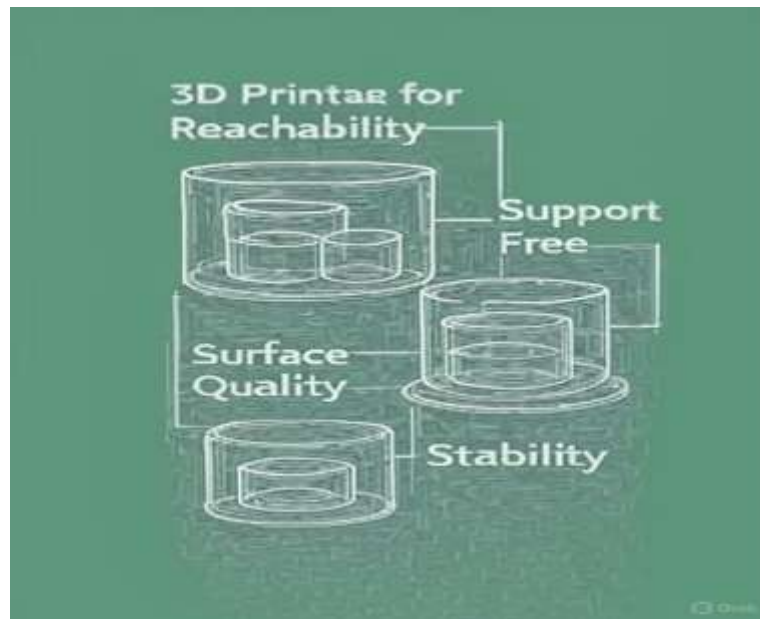


FIGURE 14:DEVELOPMENT OF CONCEPT.

Concept Development

Following the identification of these core points, the next step involves translating them into detailed functional and design requirements. These requirements stem from the core priorities, the project assignment, team input, and general engineering principles such as safety. They are categorized into mandatory “must” criteria and beneficial “wish” criteria, with design space requirements always classified as essential. The detailed requirements are outlined in the following tables.

TABLE 3:PHASE 1 – FINGER PRINTER FUNCTIONAL REQUIREMENTS

Category	Requirement Description	Status
Core Functionality	Must support all essential Tronxy X5SA Pro print functions with modular improvement.	Must
	Must print overhangs up to 65 mm at angles down to 135° without support.	Must
	Must print full 360° undercuts with ≥ 15 mm height.	Must
	Must maintain high-quality vertical surface finishes (no ripple or distortion).	Must
	Must support a minimum working volume of 200×200×200 mm.	Must
Mechanical	Additional hardware must increase the head weight by no more than 30%.	Must
	Total added weight (including all modules) must stay under 480 g.	Must
	New parts must not reduce dynamic performance or introduce backlash.	Must
	System must ensure nozzle accuracy of ± 0.05 mm or better.	Must
Material & Energy	Must be compatible with at least 6 filament types (PLA, PETG, TPU, ABS, Nylon, CF).	Must
	Extra electrical load must remain under 1.8× original power consumption.	Wish
	Hot and cold zones must be thermally isolated.	Wish
Safety	All moving elements must be guarded against human contact.	Must
	Electrical components must be insulated and protected from touch.	Must
	Must avoid any collision with printer frame or belts.	Wish
Usability	System must be installable using basic tools within 15 minutes.	Must
	Must be adaptable to a UFACTORY Lite 6 robot with ≤ 3 modified parts.	Wish
	Total added material cost (excluding motors/electronics) must be $\leq$ €180.	Wish
Durability	Should operate for at least 120 hours without failure or quality loss.	Wish
	Must successfully print a test part.	Must
	All parts should be printable via FDM or machinable using basic CNC methods.	Must

Stepper Motor Specifications:

Model: 17HS4023



FIGURE 15:NEMMA 17

<https://de.aliexpress.com/item/1005003897311523.html>

TABLE 4:MOTOR PARAMETERS

Parameter	Specification
Motor Type	NEMA 17 Stepper Motor (Model 17HS4023)
Step Angle	1.8° per step
Motor Length	Approx. 23 mm
Rated Current	1.0 A per phase
Phase Resistance	4.1 Ω
Phase Inductance	4.1 mH
Rated Voltage	DC 4.1 V
Holding Torque	Approx. 30.8 g·cm ²
Motor Weight	Approx. 132 g
Lead Wires	4 wires (bipolar)

Servo Motor Specifications:

Model: DS3218 PRO Servo Motor



FIGURE 16:SERVO MOTOR

20 kg Servo DS3218/PRO 180 270 Grad Hochgeschwindigkeitsservo Wasserdichtes Metallgetriebe Digitalservo RC-Servo für RC-Cars im Maßstab 1/8 1/10 - AliExpress 26

TABLE 5:PHASE 1 – PARAMETERS AND SPECIFICATION.

Parameter	Specification
Size	40 × 20 × 40.5 mm
Weight	60 g
Gear Ratio	275:1
Control Signal	PWM (500–2500 μs)
Rotation Range	180° or 270°
Voltage	5.0 – 6.8 V
Stall Torque	21 – 23.5 kg cm
Speed (@6 V)	0.10 – 0.12 sec/60°
Stall Current	2.1 – 2.9 A
Protection	IP66 Waterproof

Concept Phase 1



FIGURE 17:HEAD ASSEMBLY

Concept Overview:

- The design features two arms mounted on two rotational joints.
- The upper joint allows rotation about the vertical axis.
- The lower joint allows rotation about the horizontal axis.
- A 90° inclined nozzle provides full reachability in the workspace with only two joints, keeping the design simple and compact.

Drive and Movement:

- Both joints are powered by standard stepper motors with gears, ensuring high precision and low cost.
- Filament and cables pass through the cup holding the upper gear, enabling nearly endless 360° rotations for smooth operation.

Design and Assembly:

- The arm assembly can be mounted on a table bracket and secured with screws.
- Custom components are made from Aluminium 6061 to allow easy and inexpensive manufacturing.
- The design prioritizes easy assembly and maintenance.

Phase 2 – Structural Optimization and Process Simulation

Concept Enhancement and Functional Validation

After establishing the core concept, initial system layout, and base requirements in Phase 1, the focus in Phase 2 shifts toward a more detailed development process. This phase emphasizes design refinement, functional optimization, and simulation-based validation to ensure that the concept performs reliably under realistic conditions.

Additional Requirements Introduced in Phase 2

To ensure that the prototype can meet practical expectations, the following design and functional constraints were introduced by the client:

- Minimum build space: 250 mm × 250 mm × 300 mm
- Minimum print angle capability: 90°
- Full requirement specification provided for validation
- Demonstration of printability through:
 - A reference geometry (“*Stairs*”)
 - A custom-designed geometry featuring freeform surfaces and significant overhangs

These new requirements are intended to verify system performance under real-world challenges and help identify optimization potentials in design and control.

Core Objectives and Deliverables

Phase 2 consists of a sequence of technical tasks designed to strengthen the overall solution.

These include:

- Updating the existing design to comply with the newly added functional and geometric constraints
- Carrying out topology optimization with a focus on achieving a high strength-to-weight ratio
- Completing detailed CAD modelling, including standard components, parametric constraints, and a fully structured assembly
- Conducting structural analysis to investigate stress distribution, deformation, and material performance
- Performing a transient multibody simulation of the printing process using two models:
 - The predefined sample
 - The custom-designed sampleThis simulation includes trajectory planning (Cartesian and joint space), force/torque calculations for the actuators, and validation of workspace coverage.

Design Strategy for the Custom Print Sample

The custom sample created by the team is a functional benchmark used to evaluate the system's flexibility, precision, and structural print quality. Rather than being part-specific, it is crafted to challenge the mechanism's capabilities and test the core requirements identified earlier.

The design objectives for this sample include:

- **Accessibility and maneuverability:** Showcasing the ability to reach complex overhangs and undercut features
- **Support-free printing:** Ensuring that all geometries can be fabricated without auxiliary support structures
- **Enhanced surface finish:** Achieved through controlled nozzle orientation, allowing consistent deposition angles that minimize visible layer lines
- **Mechanical reliability:** The print path is planned to avoid unfavourable layer orientations, ensuring better strength and material bonding across all regions

This comprehensive evaluation confirms the feasibility of the system and provides a strong foundation for potential improvements in subsequent development stages.

The requirements for Phase 2 remain mostly consistent with those of Phase 1, with a few small updates.

TABLE 6:PHASE II – FINGER PRINTERL REQUIREMENTS

Category	Requirement Description	Status
Core Functionality	Must support all essential Tronxy X5SA Pro print functions with modular improvement.	Must
	Must print overhangs up to 65 mm at angles down to 135° without support.	Must
	Must print full 360° undercuts with ≥ 15 mm height.	Must
	Must maintain high-quality vertical surface finishes (no ripple or distortion).	Must
	Must support a minimum working volume of 250×250×300 mm.	Must
Mechanical	Additional hardware must increase the head weight by no more than 30%.	Must
	Total added weight (including all modules) must stay under 480 g.	Must
	New parts must not reduce dynamic performance or introduce backlash.	Must
	System must ensure nozzle accuracy of ± 0.05 mm or better.	Must
Material & Energy	Must be compatible with at least 6 filament types (PLA, PETG, TPU, ABS, Nylon, CF).	Must
	Extra electrical load must remain under 1.8× original power consumption.	Wish
	Hot and cold zones must be thermally isolated.	Wish
Safety	All moving elements must be guarded against human contact.	Must

	Electrical components must be insulated and protected from touch.	Must
	Must avoid any collision with printer frame or belts.	Wish
Usability	System must be installable using basic tools within 15 minutes.	Must
	Must be adaptable to a UFACTORY Lite 6 robot with ≤ 3 modified parts.	Wish
	Total added material cost (excluding motors/electronics) must be $\leq$ €180.	Wish
Durability	Should operate for at least 120 hours without failure or quality loss.	Wish
	Must successfully print a stair model provided by clients(professors) and group example.	Must
	All parts should be printable via FDM or machinable using basic CNC methods.	Must

Concept Optimization and Structural Improvement

To ensure better mechanical stability, enhanced print accuracy, and reduced part complexity, the original design underwent critical optimization. This second-phase redesign focuses on structural refinement and mechanical robustness, guided by feedback from preliminary testing and practical integration requirements.

Frame and Support Redesign

The initial arm support structure featured a U-shaped configuration, which introduced minor flex under operational loads, as observed during initial testing. To address this, the frame was re-engineered into a straight, vertical alignment. This modification involved:

- A **direct connection between the servo motor and extruder head**, mounted along a rigid central axis.
- Introduction of a **stiffened support plate**, integrated closely with the main frame to reduce deflection and improve dimensional stability.

This change results in a more compact, robust, and collision-resistant structure, with improved force flow throughout the assembly.



FIGURE 18: PHASE 1



FIGURE 19: PHASE 2.

Bearing Integration – DN625

To enhance the rotational stability of the moving components, a **DN625 bearing** was incorporated into the revised concept. This standard bearing allows smooth rotary motion while effectively distributing loads across the axis. Key improvements include:

- **Reduced mechanical play** during printing operations, improving repeatability.
- Enhanced support at critical rotational joints to sustain dynamic forces.
- Cost-efficient solution using readily available components, without the need for high-end custom parts.

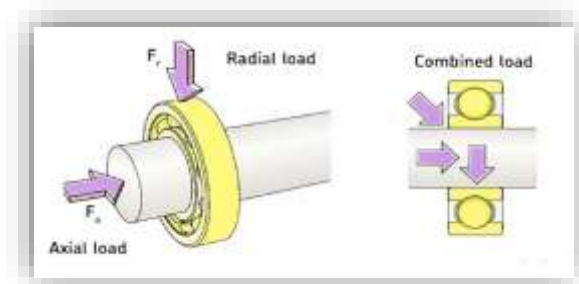


FIGURE 20: RADIAL AND AXIAL LOADS.

<https://www.lnbbearing.com/bearing-load-radial-vs-axial.html>

Improved Structural Rigidity and Print Precision

The updated configuration achieves a significant increase in stiffness due to:

- **Centralized alignment of components**, reducing overhangs and torque-induced bending.
- **Simplified load paths** that improve the handling of dynamic forces during print head movement.

Overall, the revised design is **better suited for precise and reliable printing**, particularly under demanding paths and longer print durations.

Resulting Benefits

- **Minimized deflection** during high-speed movements.
- **Improved alignment** between mechanical and nozzle axes.
- **Greater reliability and robustness**, especially under accidental disturbances or rapid motion.
- Optimized for the **working volume of $225.39 \times 270.63 \times 225.23$ mm**, maximizing usage of the available space.

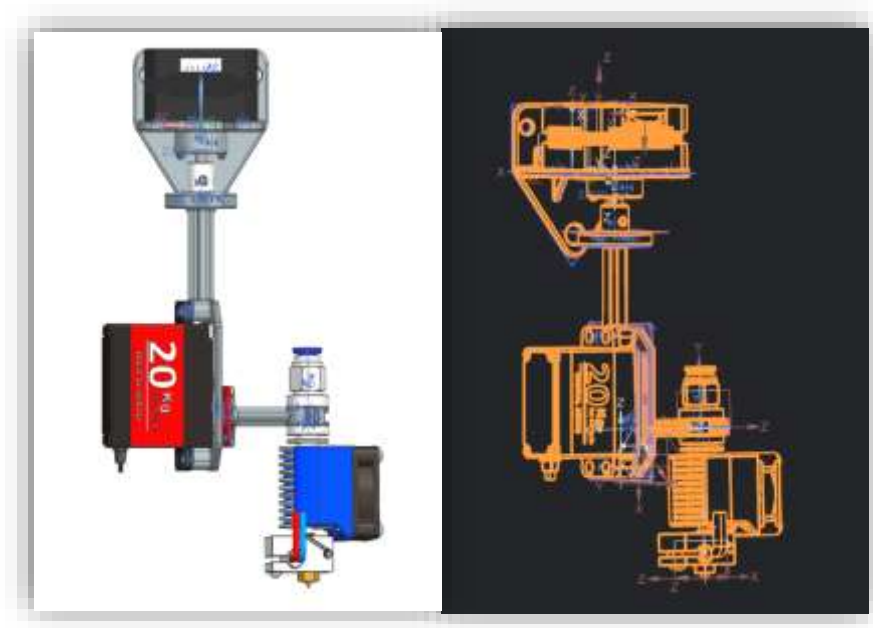


FIGURE 21:OPTIMISED ASSEMBLY

Component-Level Optimization and Simulation Strategy

Once the overall concept was finalized, the next crucial phase focused on refining each component through simulation and optimization. Utilizing advanced simulation tools, each group member was assigned clear responsibilities to ensure detailed analysis and performance enhancement of the design.

Although the project comprises four main components—the **support plate**, **main frame**, and **two clamps**—the workload was evenly distributed among the four team members. This included deeper subdivision of tasks within individual components, such as strength analysis, mounting evaluation, material selection, and assembly constraints.

- **Support Plate:** Evaluated for structural integrity, load distribution, and attachment points for moving elements.
- **Main Frame:** Optimized for stiffness, weight efficiency, and housing compatibility.
- **Two Clamps:** Focused on secure gripping, adaptability to varying geometries, and resistance to mechanical fatigue.

This systematic approach ensured that every aspect of the design was critically assessed and enhanced, with each member taking full ownership of their respective area of analysis.

Support Plate – Design Optimization (Phase 2 vs Phase 1)

The support plate has undergone a significant transformation from its initial design. One of the most notable improvements is the integration of a bearing housing into the central part of the plate. In the earlier version, no provision was made for supporting a bearing, which meant that all radial and axial loads from the motor shaft were directly transferred to the plate and motor mount. This not only risked misalignment and increased wear but also limited the system's ability to handle dynamic forces effectively. By introducing a dedicated bearing seat, the new design allows for decoupling of loads between the motor and the moving platform, enhancing both stability and motor lifespan.

Major advancement is the application of topology optimization, which is clearly visible in the form of strategically placed material cutouts and ribs. These optimizations reduce the overall mass of the component by removing material from areas experiencing minimal stress, while reinforcing zones that carry higher loads-particularly around the bolt mounting points and the bearing interface. As a result, the new plate design provides higher stiffness-to-weight ratio, improving the dynamic response of the printer's moving assembly.

To optimize the CAD components, Altair Inspire software was employed. Through multiple iterations of topology optimization, the design was refined and finalized, achieving a 39% reduction in mass while maintaining high structural stiffness and load-bearing capacity. The applied force conditions, constraint locations, and defined design space are illustrated in the figures below.

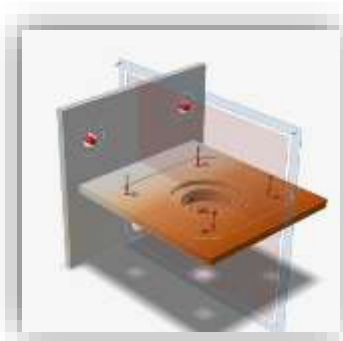


FIGURE 22:BEFORE TOPOLOGY



Figure 23: After topology

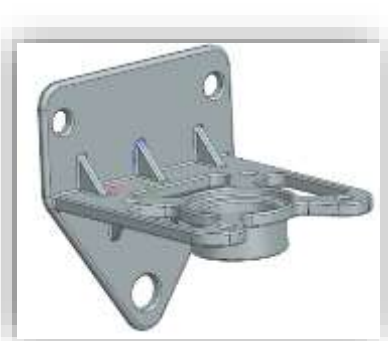


FIGURE 24:OPTIMIZED CAD MODEL

The initial weight of the support plate was 46.3 grams. After applying topology optimization, the weight was successfully reduced to 28.2 grams, resulting in a 39% reduction in mass. This significant weight savings contributes to improved performance, reduced inertia, and greater overall efficiency of the assembly.

Structural Performance Evaluation

The displacement and stress results, as illustrated in the figures below, confirm the structural integrity of the optimized support plate. Under the specified loading conditions, both the maximum displacement and stress values remain well within the material's allowable limits, ensuring the part's safety and reliability during operation.

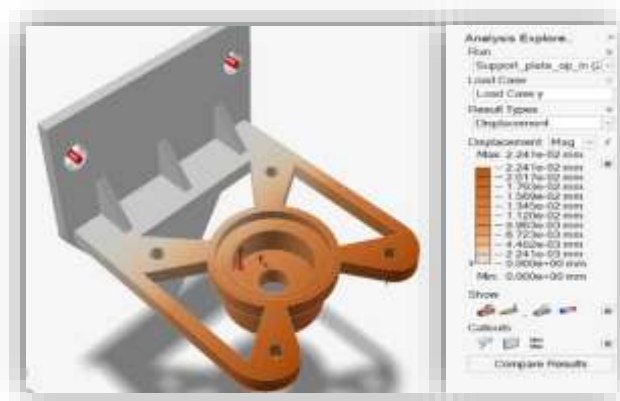


FIGURE 25:STRESS ANALYSIS.

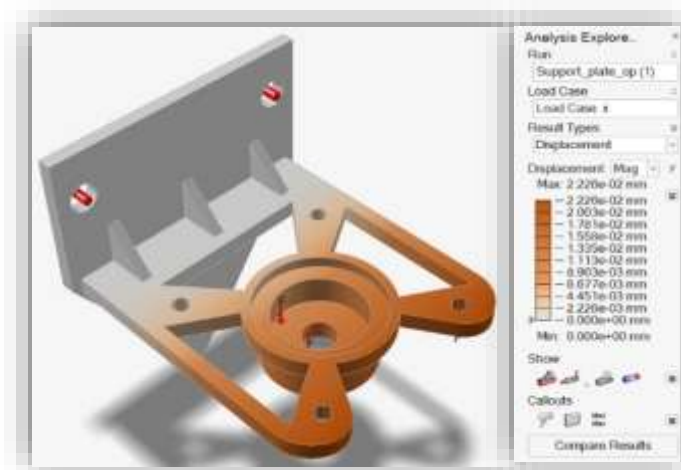


FIGURE 26:LOAD CASE X.

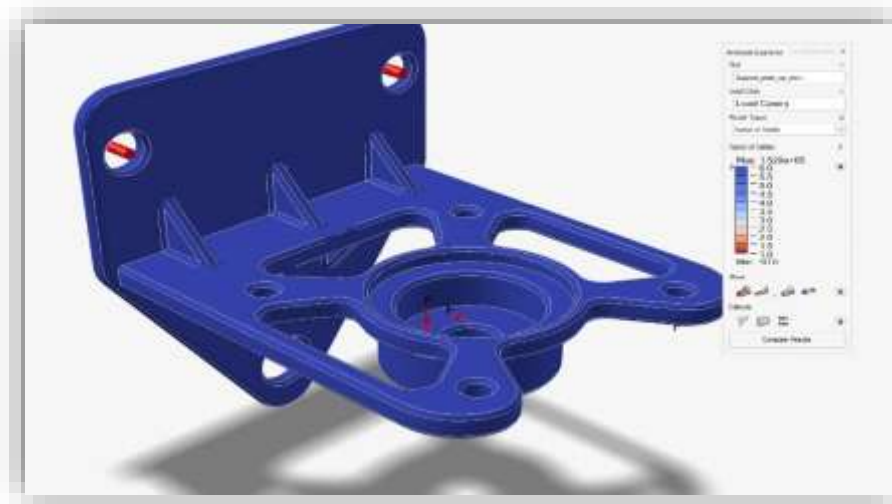


FIGURE 27:LOAD CASE Y

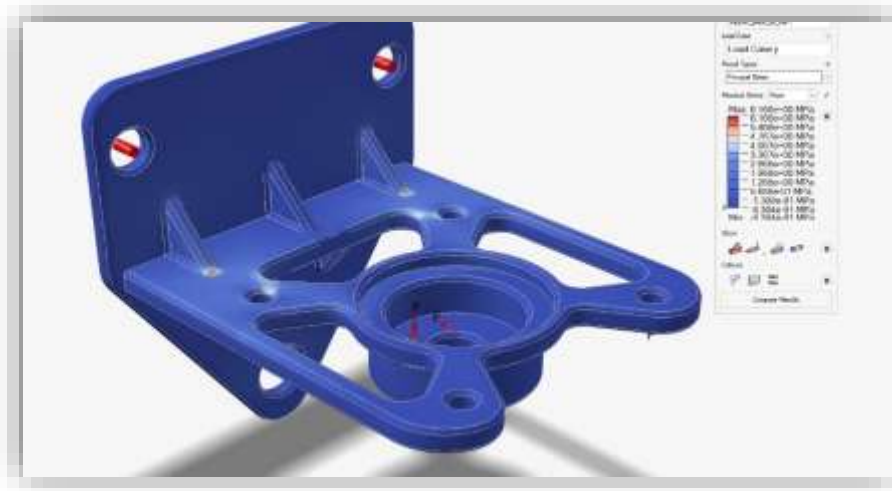


FIGURE 28:PRINCIPAL STRESS.

Main Frame

The main frame has undergone a significant geometric transformation from its initial U-shaped layout to a more streamlined straight design in the Phase 2 revision. The original U-shaped configuration, while structurally rigid, introduced manufacturing complexity, uneven stress distribution, and an inefficient use of material. Moreover, it posed limitations in terms of direct integration with other components such as the support plate and clamps.

In Phase 2, the redesign approach prioritized functional alignment and load-oriented design. The new straight configuration not only simplifies the geometry but also allows for better

symmetry in force transmission and easier manufacturability. This new structure establishes a clear, linear connection path between subcomponents, improving the assembly process and enhancing mechanical integrity.

To further refine the structural performance, topology optimization was applied using Altair Inspire. The aim was to remove unnecessary material while preserving stiffness and strength in key load-bearing areas. The resulting design features thin walls, internal ribs, and voids, which follow the natural flow of stress through the part. These changes improve the overall mass-to-stiffness ratio and allow for better dynamic response under operational loads such as motor torque and assembly weight.

FIGURE 29:MAIN FRAME MODIFICATION AND ANALYSIS



The initial weight of the main frame was 31.2 grams. Following multiple iterations of topology optimization with a target mass retention of around 55%, the final optimized design was reduced to 17.5 grams. This equates to a 44% weight reduction, significantly lowering the overall system mass. Such a reduction plays a crucial role in improving the dynamic response of the 3D printer assembly, reducing inertial loads, and enhancing energy efficiency during rapid positioning and motion cycles.

Structural Performance Evaluation



Figure 30: Load case X



FIGURE 31: LOAD CASE Y, DISPLACEMENT



FIGURE 32: LOAD CASE X, PRINCIPAL STRESS

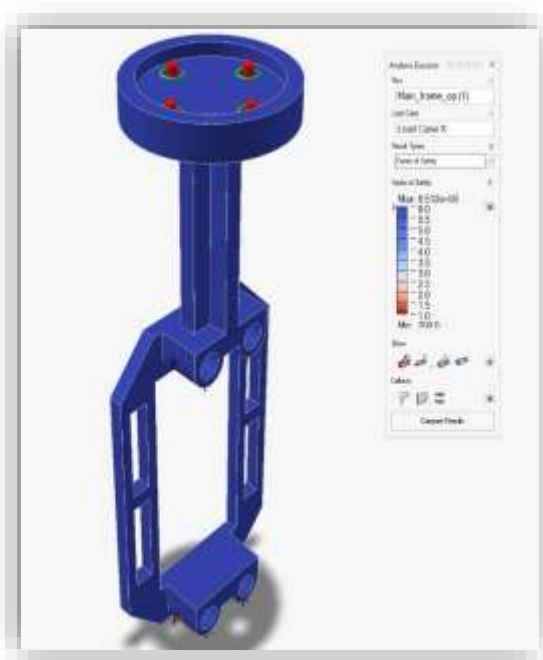


FIGURE 33: LOAD CASE Y, PRINCIPAL STRESS

Displacement and stress analysis

BIG CLAMP

In the initial design, the large clamp had a rectangular and bulky geometry, which lacked tailored load paths and introduced unnecessary weight. It was simply mated to the main frame without any specific load channeling strategy. In the updated version, the clamp has undergone topology optimization and now features streamlined contours that more effectively align with the straightened and optimized main frame. The previously rigid U-shape main frame has now been replaced with a lighter and straighter structure, allowing the clamp's geometry to adapt accordingly.

The mounting region of the clamp—where it connects to the main frame—has been thickened selectively, reinforcing high-stress areas. The updated contact surfaces are now better matched with the revised main frame, leading to more stable fixation and minimizing micro-shifts during operation. Furthermore, the mass reduction in both the main frame and the clamp leads to a cumulative improvement in system dynamics, enhancing acceleration response and reducing inertia during quick direction changes.

From a manufacturing point of view, the new clamp is more suitable for lightweight machining or 3D printing, offering smoother curves and reduced material usage. Functionally, it enables stronger and more reliable clamping, better alignment with the main frame, and improved longevity under repetitive mechanical stresses. This redesign ensures that the clamp does not act as a structural weak point, but rather complements the performance and durability of the entire assembly.

Structural Performance Evaluation

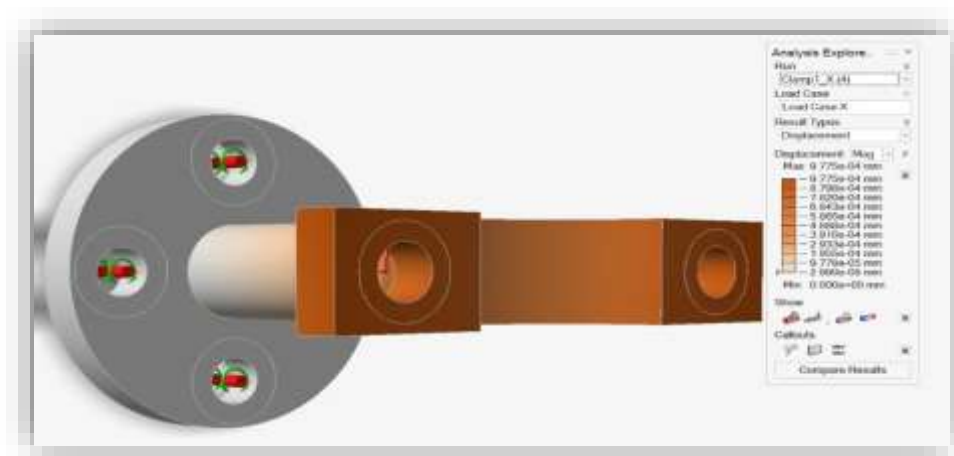


FIGURE 34: CLAMP ANALYSIS

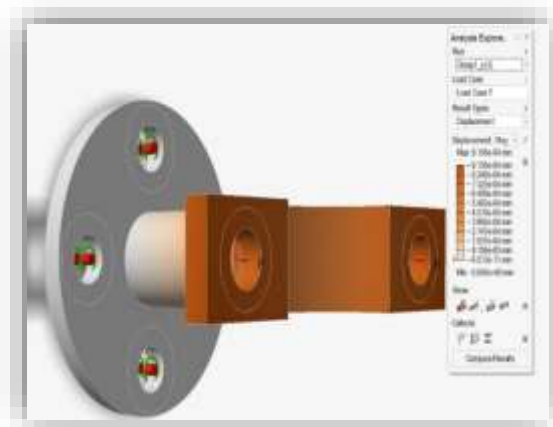


FIGURE 35: CLAMP LOAD CASE Y

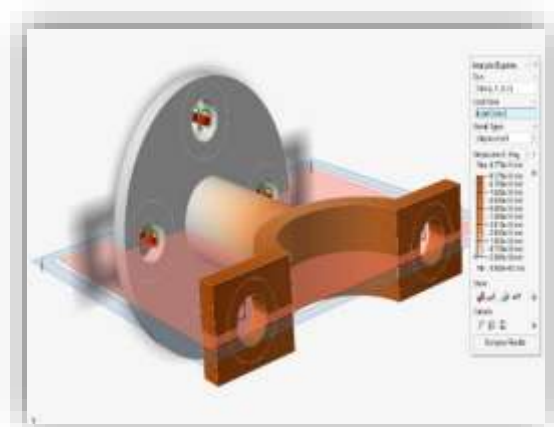


FIGURE 36: CLAMP LOAD CASE 2 DISPLACEMENT.

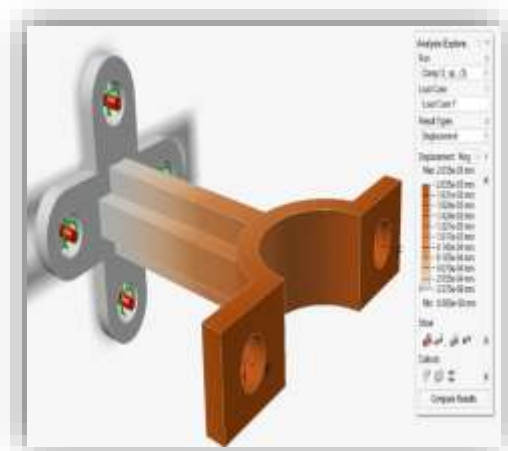


FIGURE 37 :CLAMP LOAD CASE Y AFTER OPTIMISED

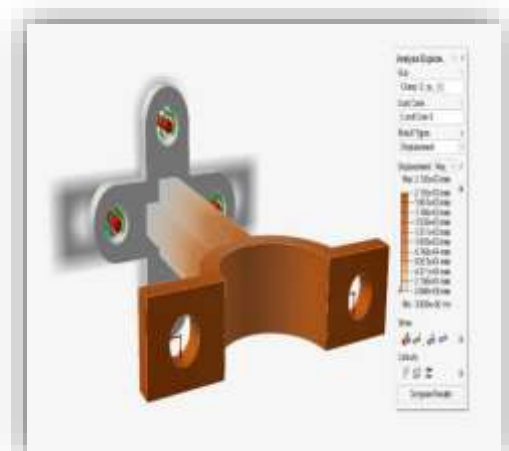


FIGURE 38 :CLAMP LOAD CASE X



FIGURE 39: SMALL CLAMP TOPOLOGY OPTIMIZATION

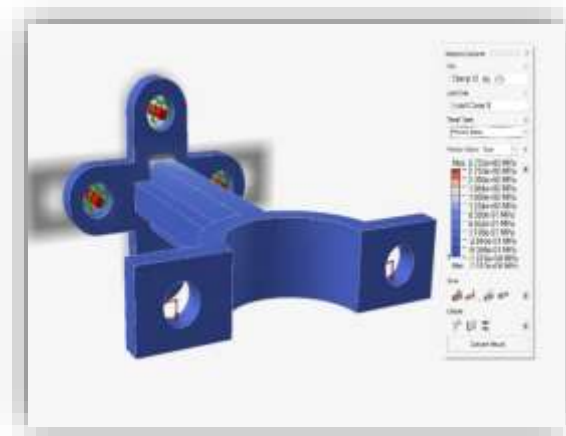


FIGURE 40 :CLAMP LOAD CASE X MISSES STRESS.

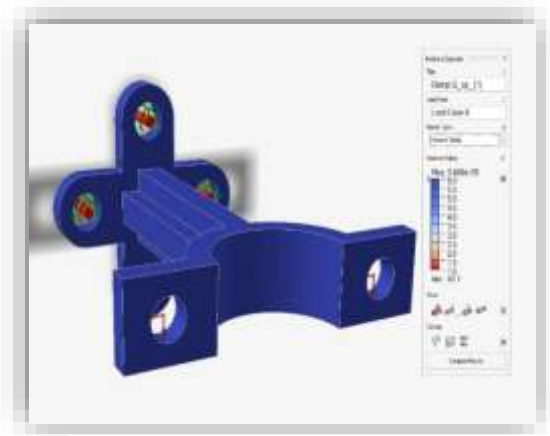


FIGURE 41 :CLAMP LOAD CASE Y MISSES STRESS.

The initial weight of the large clamp was 4.1 grams. After applying topology optimization with a focus on structural efficiency and minimal mass retention, the final design was reduced to 3.1 grams. This results in a 24% weight reduction, which contributes meaningfully to the overall system's dynamic performance. The reduction in clamp mass helps decrease inertial resistance during high-speed print head movements, leading to improved responsiveness, lower energy consumption, and reduced stress on mounting components.

SMALL CLAMP



FIGURE 42 :SMALL CLAMP.

The small clamp, originally weighing 1.0 gram, did not undergo full topology optimization due to its already minimal mass. However, significant design changes were implemented to improve functionality and structural integration with the optimized main frame.

The geometry was refined to feature smoother contours and better alignment at the mounting interface, improving fixation and load distribution. These updates helped reduce the weight slightly to 0.8 grams-a 20% decrease-while enhancing mechanical stability during dynamic motion. The redesigned clamp complements the streamlined frame and improves overall assembly performance without compromising structural integrity.

Motion View-Based Simulation of Finger Printer: Initial Results

To simulate the motion of the finger printer in Altair MotionView, a primitive model was created and later validated with the imported CAD model. As seen in the image, the CAD model consists of five rigid bodies:

1. **Support Plate** – shown in orange
2. **Main Frame** – connected below the support plate
3. **End Effector** – attached to the rotating shaft
4. **X-axis Rail** – transparent grey horizontal guide
5. **Y-axis Rail** – green horizontal guide perpendicular to X

All components were modelled to match the assembly geometry of the actual design.

Joint Configuration:

- The Support Plate is connected to the Ground through an Orientation Joint. This joint allows translation in X, Y, and Z directions but constrains all rotational DOFs (roll, pitch, yaw), enabling smooth sliding motion along the rails and vertical axis.
- The Main Frame is connected to the Support Plate using a Revolute Joint, allowing rotation about the Z-axis, simulating pivoting or turning motion of the main assembly.
- Another Revolute Joint connects the Main Frame to the End Effector, enabling the rotation
- n of the tool head (extruder) – crucial for printing curved or angled paths.

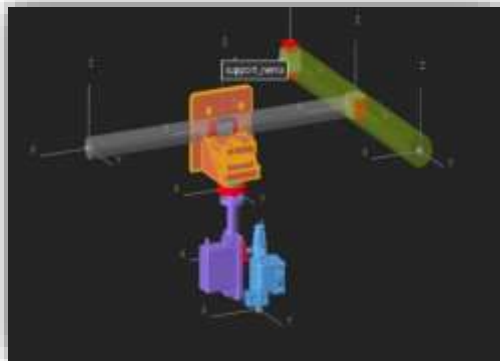


FIGURE 43: JOINT CONFIGURATION MODEL.

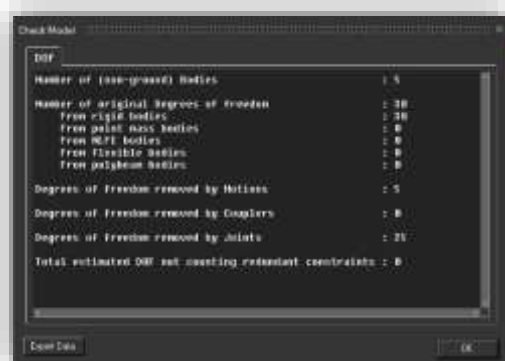


FIGURE 44: JOINT CONFIGURATION PARAMETERS.

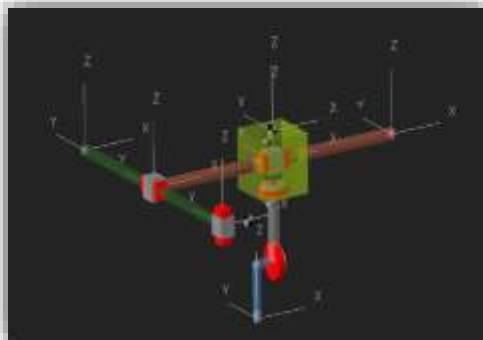


FIGURE 45: PRIMITIVE MODEL.

Degrees of Freedom (DOF) Calculation:

As per the DOF analysis report (see earlier image):

- Total rigid bodies (excluding ground): 5
- Each rigid body contributes 6 DOF →

$$5 \text{ bodies} \times 6 = 30 \text{ original DOF}$$

Constraints applied:

- Joints removed: 25 DOF
- Motions removed: 5 DOF
- Total DOF removed = 30

Final estimated DOF = 0

The model is fully constrained, with no redundant or conflicting motions.

Motion Strategy for Staircase Printing

The staircase model was printed by generating motion paths directly in Motion View. The geometry was measured and the motion definitions were created using the obtained values. The tool nozzle was controlled by prescribing translational and rotational motions through STEP5 functions, ensuring smooth velocity transitions without sharp discontinuities.

Translational Motions

X-direction motion was defined as a sequence of STEP5 functions to achieve alternating positive and negative displacements, effectively tracing the staircase path along the horizontal axis:

STEP5(TIME, 0.0, 0, 1.0, 0.15) + STEP5(TIME, 2.0, 0, 3.0, -0.15) +
STEP5(TIME, 4.0, 0, 5.0, 0.15) + STEP5(TIME, 6.0, 0, 7.0, -0.15) +
STEP5(TIME, 8.0, 0, 9.0, 0.15) + STEP5(TIME, 10.0, 0, 11.0, -0.15) +
STEP5(TIME, 12.0, 0, 13.0, 0.15) + STEP5(TIME, 14.0, 0, 15.0, -0.15) +
STEP5(TIME, 16.0, 0, 17.0, 0.15) + STEP5(TIME, 18.0, 0, 19.0, -0.15) +
STEP5(TIME, 22.5, 0.000, 24.5, 0.150) + STEP5(TIME, 30.5, 0.000, 31.5, -0.150) +
STEP5(TIME, 34.5, 0.0, 35.5, 0.15).

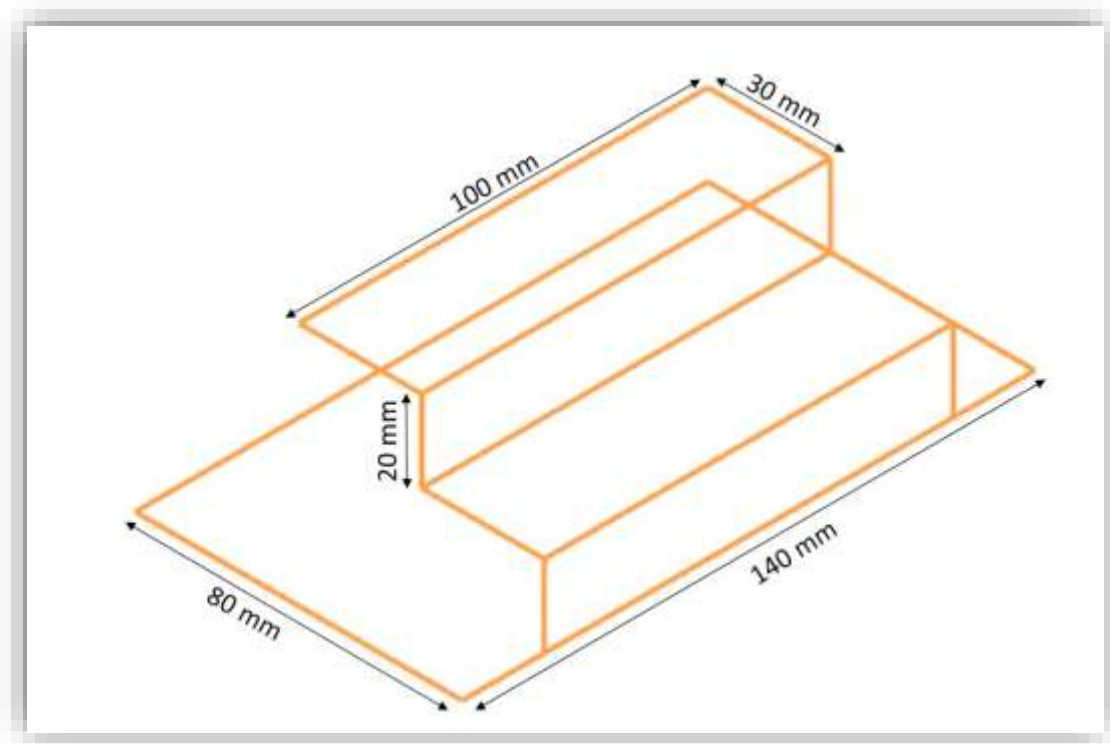


FIGURE 46 :STAIR DIMENSIONS

Y-direction motion was introduced for vertical steps using smaller displacement values:

```
STEP5(TIME, 19.5, 0.000, 20.5, -0.070) +
STEP5(TIME, 24.5, -0.070, 25.5, 0.000) +
STEP5(TIME, 28.5, 0.000, 29.5, -0.070) +
STEP5(TIME, 33.5, -0.070, 34.5, -0.140) +
STEP5(TIME, 35.5, -0.140, 36.5, -0.070.
```

Y-direction motion represented the layer height increments. A continuous sequence of STEP5 functions was applied to gradually raise the nozzle:

```
STEP5(TIME, 1.0, 0.000, 1.5, 0.003) + STEP5(TIME, 3.0, 0.003, 3.5, 0.006) +
STEP5(TIME, 5.0, 0.006, 5.5, 0.009) + STEP5(TIME, 7.0, 0.009, 7.5, 0.012) +
STEP5(TIME, 9.0, 0.012, 9.5, 0.015) + STEP5(TIME, 11.0, 0.015, 11.5, 0.018) +
STEP5(TIME, 13.0, 0.018, 13.5, 0.021) + STEP5(TIME, 15.0, 0.021, 15.5, 0.024) +
STEP5(TIME, 17.0, 0.024, 17.5, 0.027) + STEP5(TIME, 19.0, 0.027, 19.5, 0.030) +
STEP5(TIME, 26.5, 0.030, 27.5, 0.027) + STEP5(TIME, 27.5, 0.027, 28.5, 0.030) +
```


STEP5(TIME, 29.5, 0.000, 30.5, 0.030) + STEP5(TIME, 31.5, 0.030, 32.5, 0.000) +
STEP5(TIME, 32.5, 0.000, 33.5, 0.030).

Rotational Motions

Rotation about the **X-axis** was defined for a single motion event:

STEP5(TIME, 20.5, 0, 22.5, DTOR (90))

By applying this scripted sequence of translational and rotational STEP5 motions, the staircase model was successfully traced and printed in Motion View. The use of STEP5 instead of STEP ensured smoother acceleration and deceleration profiles, avoiding sharp transitions in velocity.

Results

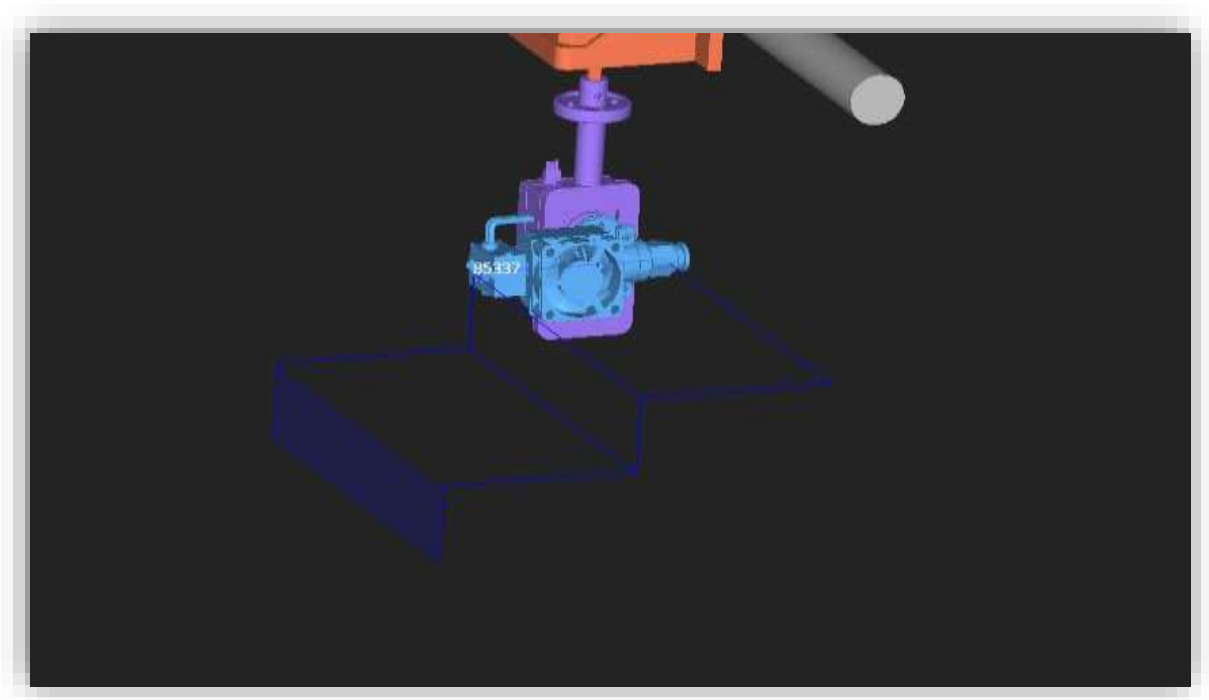


FIGURE 47: STAIR CASE RESULT.

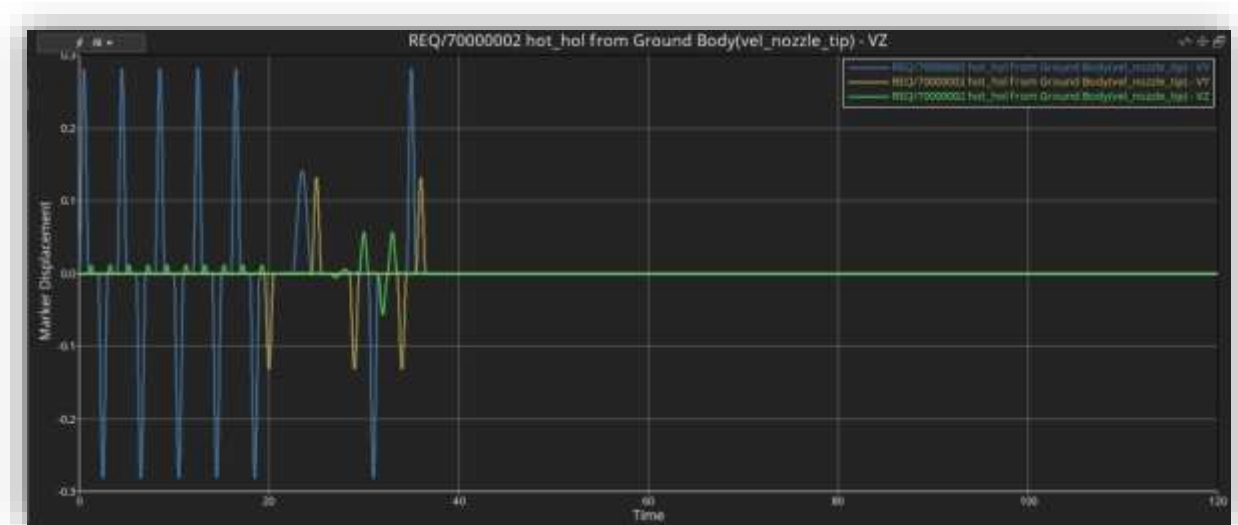


FIGURE 48: MARKER DISPLACEMENT V/S TIME VZ COMPARISON.

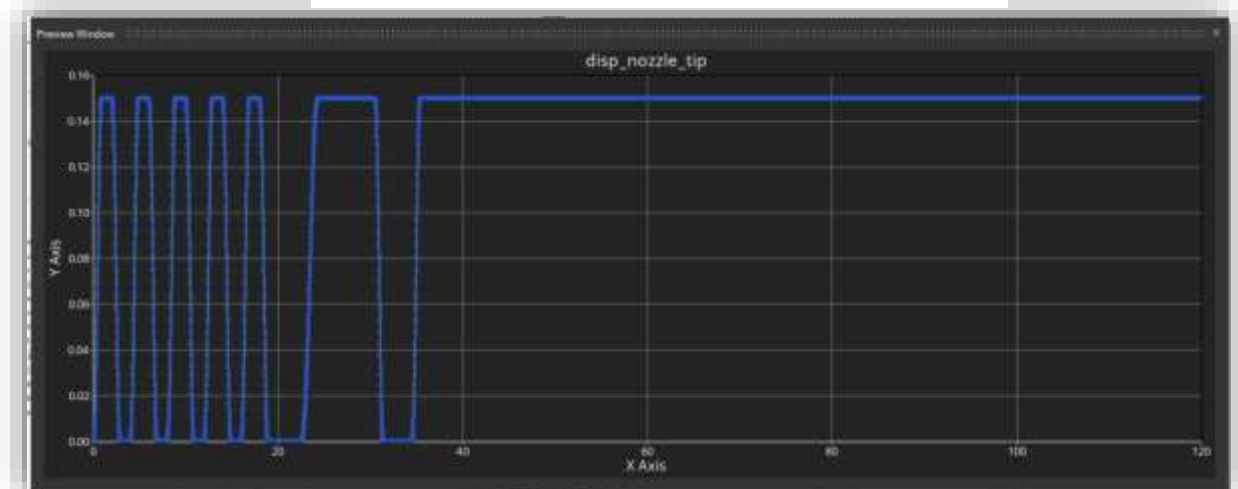


FIGURE 49 :X-AXIS V/S Y AXIS.

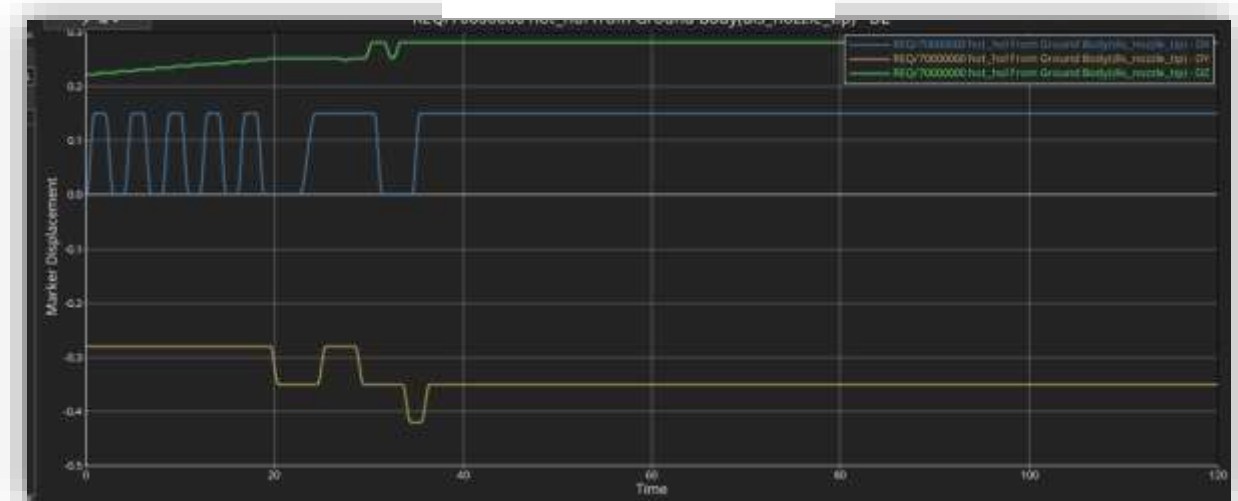


FIGURE 50: MARKER DISPLACEMENT V/S TIME.

Phase 3 – Final Validation of the Developed Concept

Calculations

Lower motor

For a point mass at a distance r from the axis of rotation:

$I = mr^2$, r is the radius of gyration obtained in CAD model (47.86 mm)

m is the mass of the lower arm and all the attached component (1144grams)

$$I = 0.1144\text{kg} * (47.86 \text{ mm}^2)$$

$$I = 0.00026 \text{ kg.m}^2$$

The Angular Acceleration (α)

Angular acceleration is the rate of change of angular velocity. If you need the arm to reach a certain angular velocity ω from rest in time t . Here we use an acceleration of $1500\text{mm} / \text{s}^2$.

$\alpha = a / r$, r is the length of the lower arm 85mm and a is the linear acceleration.

$$\alpha = 1500 \text{ mm} / \text{s}^2 / 47.86\text{mm} = 31.34\text{rad} / \text{s}^2$$

The Torque for Angular Acceleration (T_{acc})

Using Newton's second law for rotation:

$$T_{acc} = I\alpha = 0.00026 * 31.34\text{rad} / \text{s}^2$$

$$T_{acc} = 0.8\text{Ncm}$$

Gravitational Torque

If the arm segment is moving in a plane where gravity affects it, you need to calculate the torque due to gravity. For a rod of length L and mass m rotating about one end:

$$T_{\text{gravity}} = m * g * r * \sin(90)$$

$$T_{\text{gravity}} = 0.1144 \text{ kg} * 9.81 \text{ m/s}^2 * 0.048 \text{ m} \sin(90)$$

$$T_{\text{gravity}} = 5.3 \text{ Ncm}$$

where g is the acceleration due to gravity and θ is the angle the rod makes with the vertical.

Sum Up the Torques

The total required torque is the sum of the torques needed to overcome inertia, gravity, and any resistive forces:

$$T_{\text{total}} = T_{\text{acc}} + T_{\text{gravity resistive}}$$

$$T_{\text{tot}} = 0.8 \text{ Ncm} + 5.3 \text{ Ncm}$$

$$T_{\text{lower motor}} = 6.1 \text{ Ncm}$$

Upper Motor

For a point mass at a distance r from the axis of rotation:

$$I = m r^2 \quad r \text{ is the radius of gyration obtained in CAD model (42.34 mm)}$$

m is the mass of all the components driven by the upper motor and all the attached component (0.2197 Kg)

$$I = 0.2197 \text{ kg} * (42.34 \text{ mm}^2)$$

$$I = 0.0002 \text{ kg.m}^2$$

The Torque for Angular Acceleration (T_{acc})

Using Newton's second law for rotation:

$$T_{\text{acc}} = I \alpha$$

$$T_{\text{acc}} = 0.0002 \text{ kg.m}^2 * 35.42 \text{ rad/s}^2$$

$$T_{\text{acc}} = 7 \text{ Ncm}$$

Gravitational Torque

The mass of the arm is being hold by the motor along the z axis

$$T_{\text{gravity}} = m * g * r * \sin(\theta)$$

$$T_{\text{gravity}} = 0.2197 \text{ kg} * 9.81 \text{ m/s}^2 * 0.042 \text{ m} \sin(90)$$

$$T_{\text{gravity}} = 9 \text{ Ncm}$$

where g is the acceleration due to gravity and θ is the angle the rod makes with the vertical.

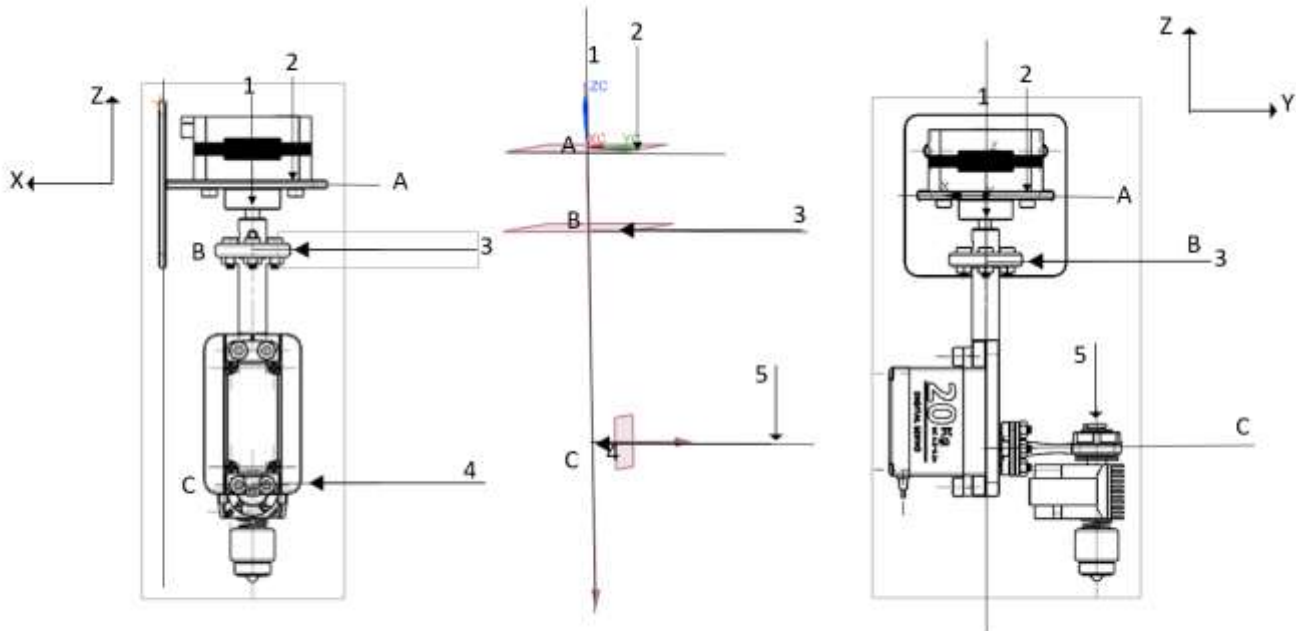
Sum Up the Torques

The total required torque is the sum of the torques needed to overcome inertia, gravity, and any resistive forces:

$$T_{\text{total}} = T_{\text{acc}} + T_{\text{gravity}} + T_{\text{resistive}}$$

$$T_{\text{tot}} = 7 \text{ Ncm} + 9 \text{ Ncm}$$

$$T_{\text{upper motor}} = 16 \text{ Ncm}$$



Shear Force and Bending Moment Calculations

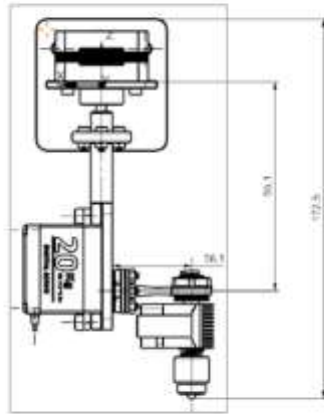


FIGURE 51: MODEL DIMENSION

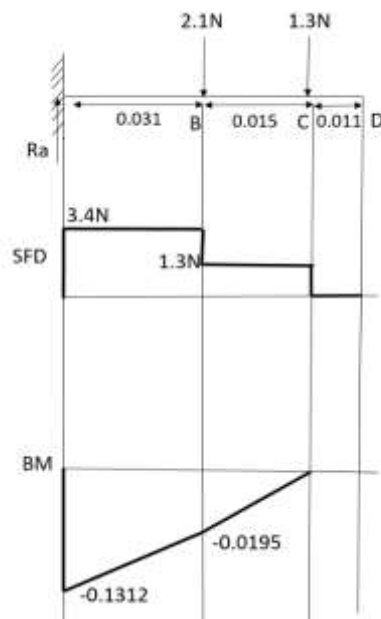


FIGURE 52: SHEAR FORCE AND BENDING MOMENT ON SUPPORT PLATE

Bending Moment and shear force on the support plate

Assume sum of Upward forces is equal to sum of downward forces $\sum \uparrow = \sum \downarrow$

$$Ra = 2.1 + 1.3 = 3.4N$$

SFD

$$Ra = 3.4N$$

$$Fb = 3.4 - 2.1 = 1.3N$$

$$Fc = 1.3 - 1.3 = 0N$$

BM

$$\text{At } C = 0$$

$$\text{At } B = -1.3 \times 0.015 = -0.0195NM$$

$$\text{At } Ra = -2.1 \times 0.031 - 1.3 \times 0.046 = -0.1312NM$$

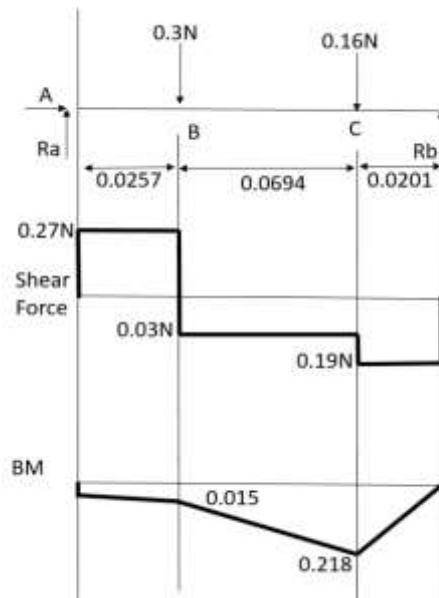


FIGURE 53: SHEAR FORCE AND BENDING MOMENT ON MAIN FRAME.

Component-Based Load Description

Point C: The 1.3 N load at point C represents the weight of a NEMA 17 motor.

Point B: The 2.1 N load at point B accounts for the combined weight of the bearing, flange, mainframe, servo, hotend, and hotend holder. This represents the total force exerted by the assembled components on the supporting structure.

Bending Moment and shear force on the Main Frame

Assume sum of Upward forces is equal to sum of downward forces $\sum \uparrow = \sum \downarrow$

$$Ra + Rb = 0.16 + 0.3 = 0.46N$$

$$0.3 \times 0.0257 + 0.16 \times 0.0951 + (-Rb \times 0.1152) = 0$$

$$\therefore Rb = 0.19N$$

$$Ra = 0.27N$$

SFD

$$Ra = 0.27N$$

$$Fb = 0.27 - 0.3 = -0.03N$$

$$Fc = -0.03 - 0.16 = 0.19N$$

$$Rb = 0.19 - 0.19 = 0N$$

BM

$$\text{At } Rb = 0.19 \times 0.1152 = 0.0218Nm$$

$$\text{At } C = 0.16 \times 0.0951 = 0.015Nm$$

The shear and bending behaviour of the supporting beam was analysed considering the dynamic loads generated by the moving components. The forces at points B and C were calculated using the component masses and an acceleration of 1500 mm/s^2 resulting in 0.3 N and 0.16 N

Bending Moment and shear force on the Hotend Mount:

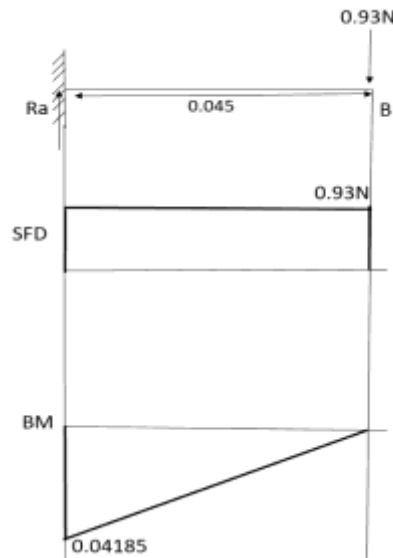


FIGURE 54: SHEAR FORCE AND BENDING MOMENT ON HOT END.

Assume sum of Upward forces is equal to sum of downward forces $\sum \uparrow = \sum \downarrow$

$$Ra = 0.93$$

SFD

$$Ra = 0.93 \text{ N}$$

$$Fb = 9.93 \text{ N}$$

BM

$$\text{At } B = 0$$

$$\text{At } Ra = -0.93 \times 0.045 = -0.04185 \text{ Nm}$$

The **0.93 N load** represents the weight of the **hot end**. This is a reasonable value, as a hot end assembly (including the nozzle, heater block, and heat break) is typically very light. For example, a common hot end like the E3D V6 weighs around 75 g

Final Optimization Results of CAD Components

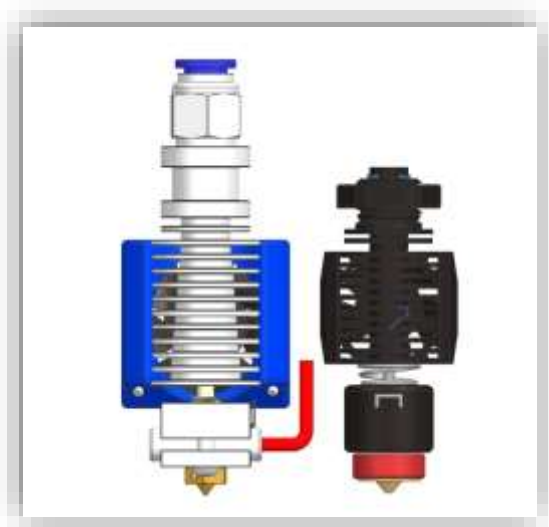


FIGURE 55: NOZZLE COMPARISON.

After the Phase 2 reviews, we decided to change the hot end. During our search, we identified the E3D Revo Micro hot end, which is more compact compared to the previous design

Due to changes in components, we repeated the topology and optimization process, which is explained below.

Support Plate:

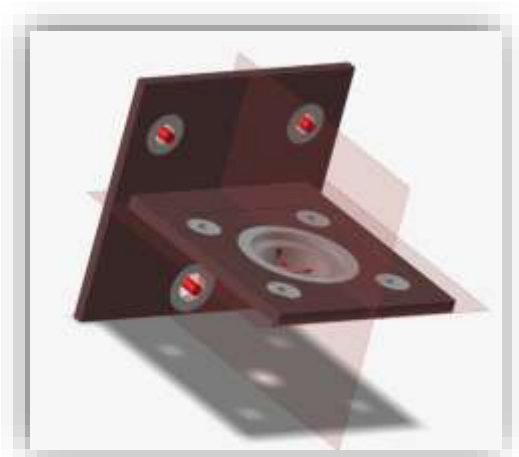


FIGURE 56: LOAD CASES ON SUPPORT PLATE

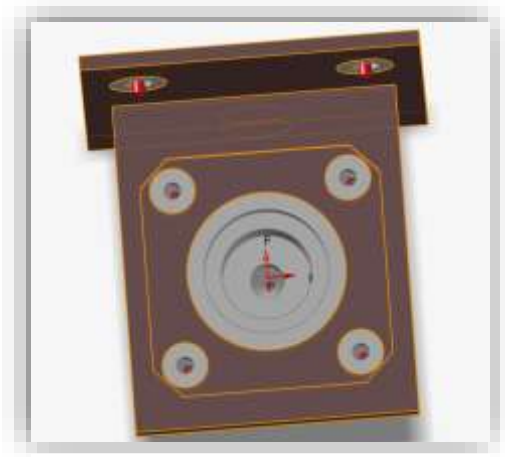


FIGURE 57: LOAD CASES ON SUPPORT PLATE.

As shown in the left image, the motor load of 0.3323 N is applied on the four mounting holes. In the right image, a 2.18 N load is applied at the central bearing seat, representing the weight of the suspended assembly excluding the motor and plate. Additionally, inertial loads of 0.54045 N are applied in both X and Y directions to account for acceleration effects. Furthermore, ribs have been added at the bottom of the plate to improve stiffness.

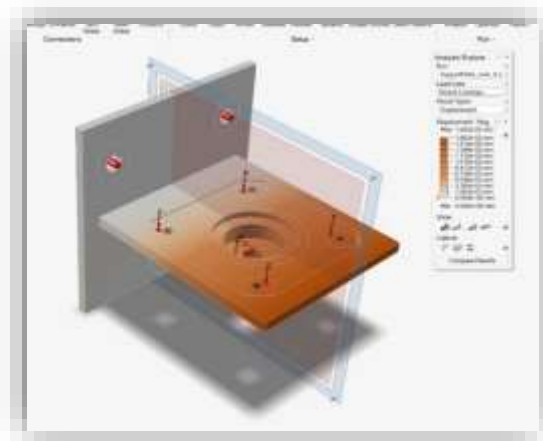


FIGURE 58: SUPPORT PLATE DISPLACEMENT

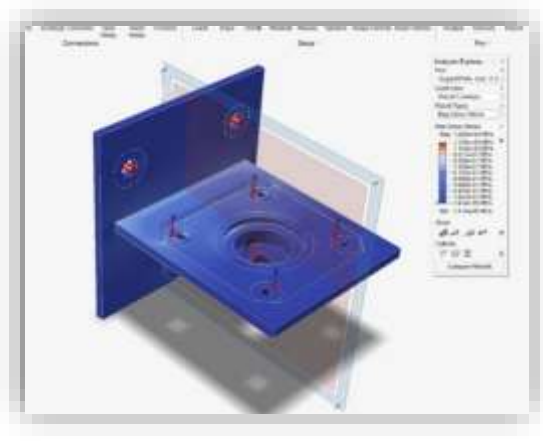


FIGURE 59: SUPPORT PLATE MAX SHEAR STRESSES.

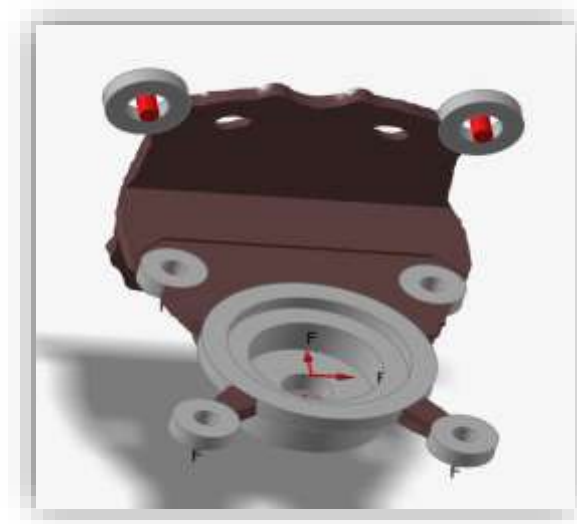


FIGURE 60: SUPPORT PLATE TOPOLOGY RESULT.

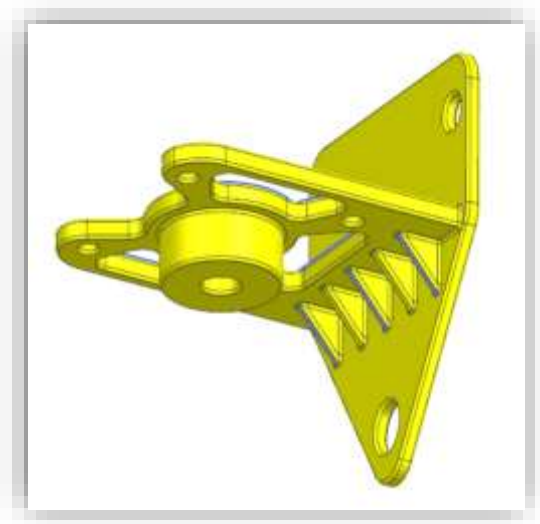


FIGURE 61: SUPPORT PLATE OPTIMISED.

The weight of the component before topology optimization was 0.0459 kg, which was reduced to 0.0283 kg after optimization. To compensate for stiffness, ribs were added at the bottom, thereby enhancing structural rigidity.

After completing the topology optimization and subsequent analysis, the results show a maximum displacement of 0.02mm under the applied loading conditions. The maximum von Mises stress observed in the structure is 3.688 MPa.

The weight of the component was reduced from 0.0459 kg before topology optimization to 0.0283 kg after optimization, representing a 38.3% reduction in mass. To maintain stiffness, reinforcing ribs were incorporated at the bottom, which enhanced the overall structural rigidity of the design.

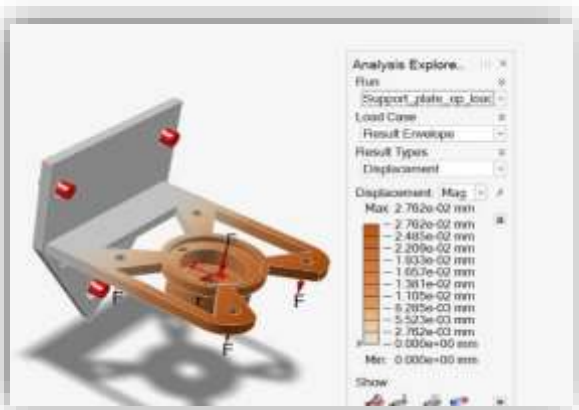


FIGURE 62:SUPPORT PLATE OPTIMISED DISPLACEMENT

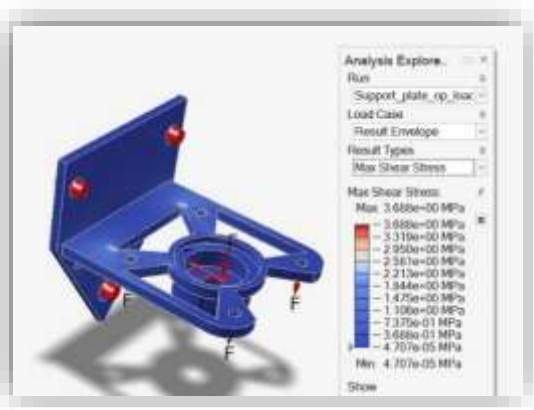


FIGURE 63:SUPPORT PLATE OPTIMISED MAX SHEAR STRESS

Main Frame:

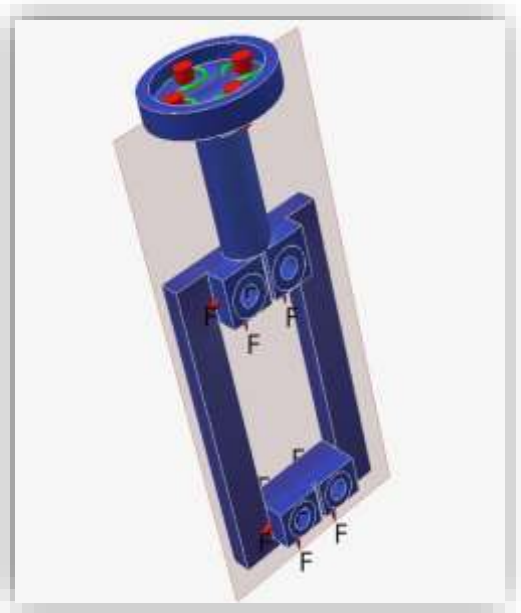


FIGURE 64:MAIN FRAME



FIGURE 65:MAIN FRAME DISPLACEMENT

In the model, the four holes at the top plate were directly bolted to the coupler and defined as fixed joints to represent the actual connection. Vertical forces of 0.4902 N were applied at each of the four bolt holes to represent the weight of the motor, hotend, and hotend mount located at the lower end of the structure. Horizontal Force Due to acceleration is 0.1716N Divide by four to each hole.

After performing the topology optimization for the main frame, the results indicated a maximum displacement of 0.06 mm and a maximum von Mises stress of 0.34 MPa.

The weight of the main frame was reduced from 0.0305 kg before topology optimization to 0.0179 kg after optimization, representing a 17.1% reduction in mass.



FIGURE 66: MAIN FRAME SHEAR STRESS

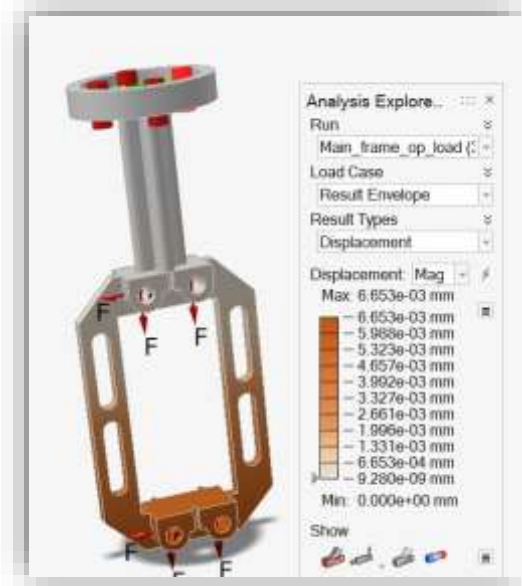


FIGURE 67: MAIN FRAME OPTIMISED DISPLACEMENT

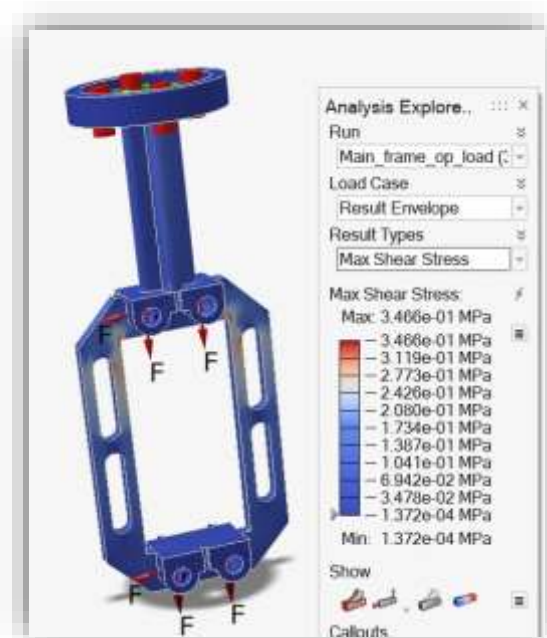


FIGURE 68: MAIN FRAME OPTIMISED SHEAR STRESS

Hot End Mount:

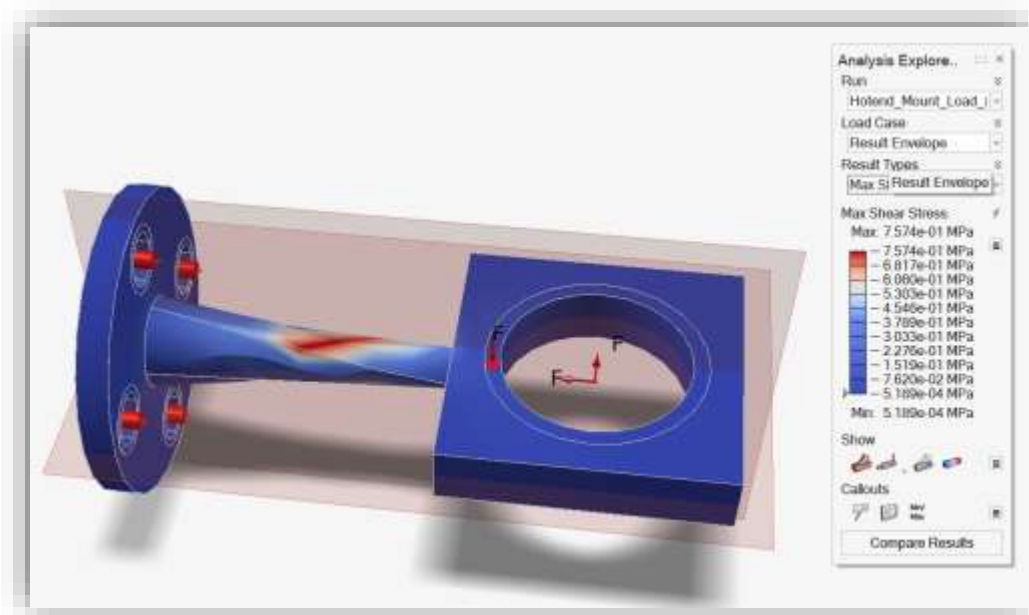


FIGURE 69: HOTEND MOUNT SHEAR STRESS

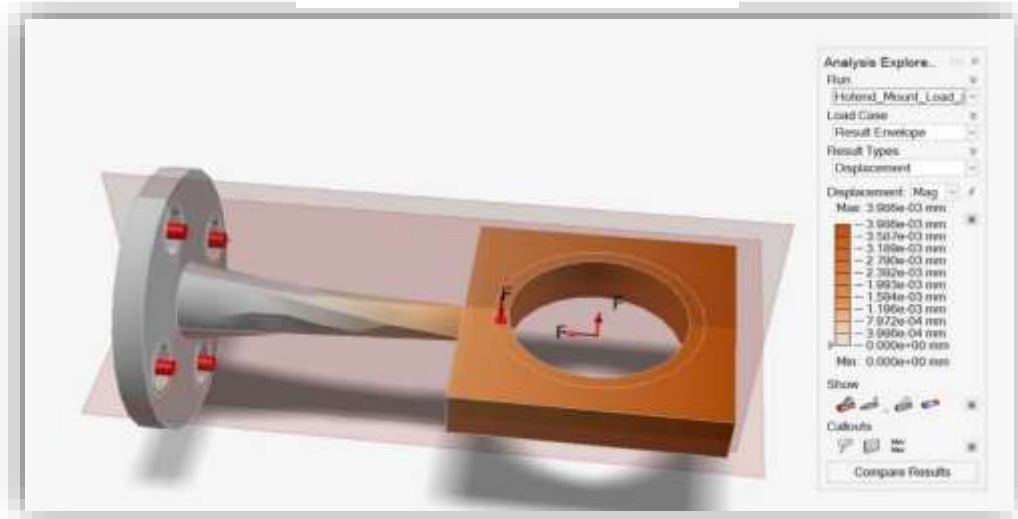


FIGURE 70: HOTEND MOUNT DISPLACEMENT

For the hot-end mount, it is fixed to the MG round coupler, which is driven by the lower motor. Therefore, the holes were modelled as torsional joints, while the hot-end load was applied at the centre of the end circle, simulating a cantilever beam. The four joints as torsional fixed joints

After performing topology and optimization, the hot end component's weight was successfully reduced from 0.0043 kg to 0.0029 kg, achieving a significant reduction of nearly 33%. This weight reduction was accomplished without compromising structural integrity, thereby improving overall efficiency and material utilization in the design.

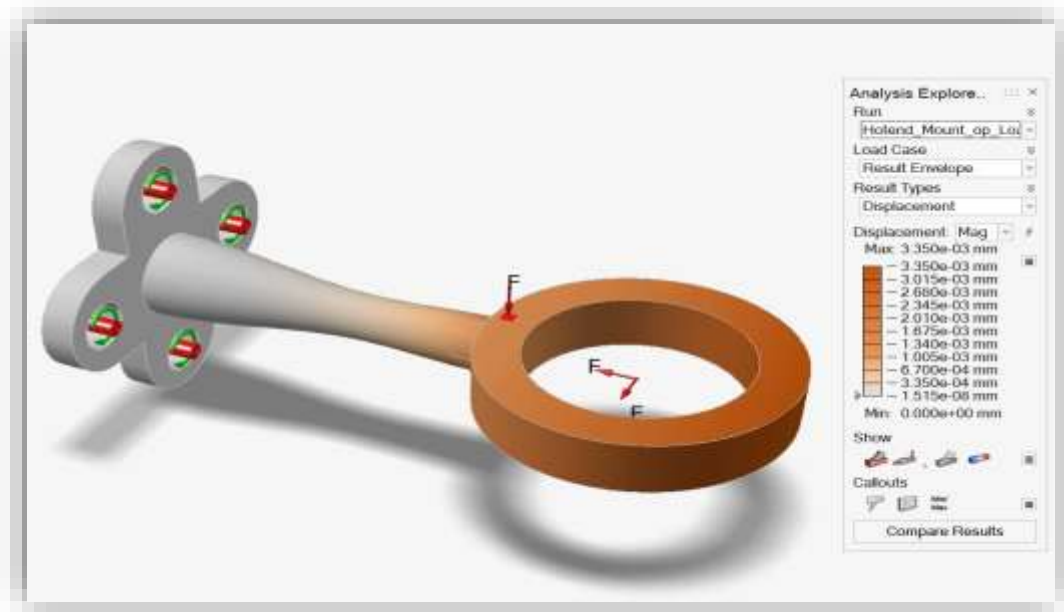


FIGURE 71: HOTEND MOUNT OPTIMISED DISPLACEMENT

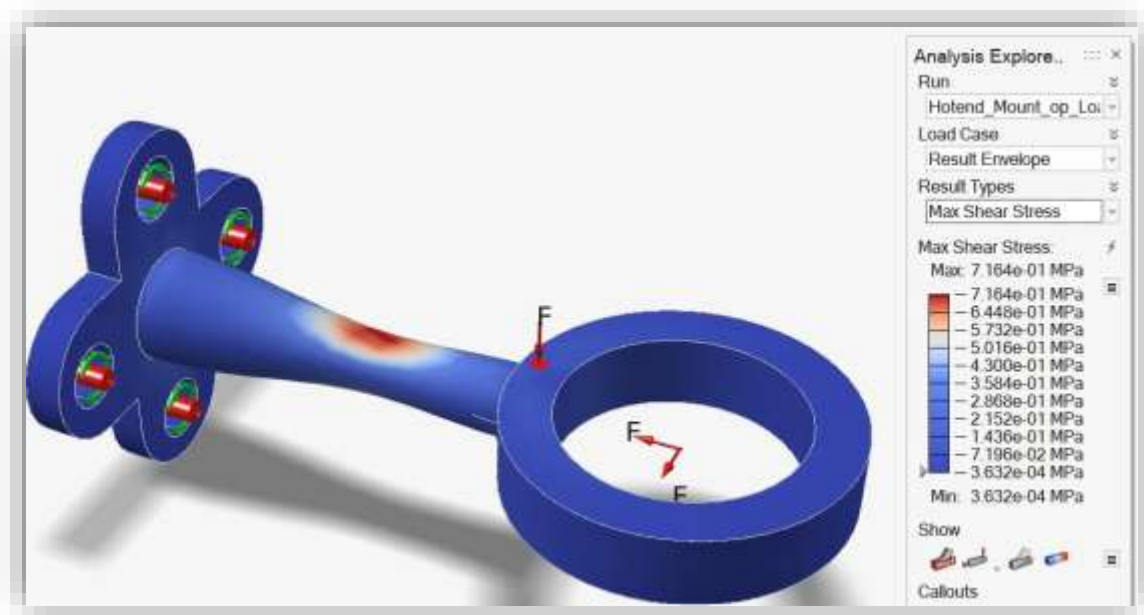


FIGURE 72: HOTEND MOUNT OPTIMISED SHEAR STRESS

Skeleton Assembly:

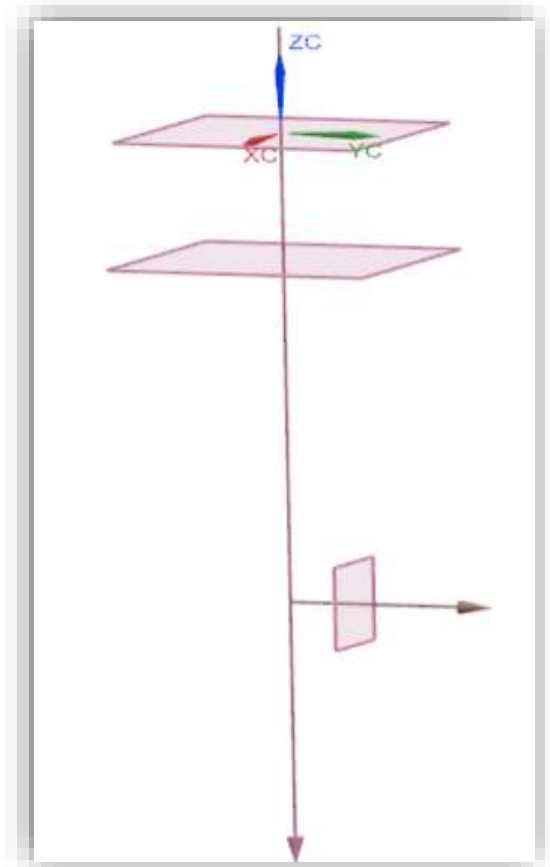


FIGURE 73: SKELETON VIEW WITH DATUM PLANES



FIGURE 74: SKELETON VIEW WITH ASSEMBLY

Multibody Simulation in Motion View: 3D Printing Parameters

Overview

This simulation models the motion parameters of a 3D printer using constant velocity and variable acceleration settings based on real-world 3D printing standards. The model we considered print speed of 60 mm/s and an acceleration of 1500 mm/s² to reflect a balanced configuration suitable for typical FDM (Fused Deposition Modelling) printing scenarios.

Material Considerations for Print Speed

The type of material being used also influences what a good printing speed is. Here are some general guidelines for various materials:

PLA: While the recommended speed for everyday high-quality results is around 50–60 mm/s, PLA is particularly well-suited for high-speed printing, enabling users to significantly increase the upper limit to more than 250 mm/s.

ABS: Print at 60 mm/s for optimal quality, with the same speed as the upper limit.

PETG: A typical print speed of 50 mm/s works well, with an upper limit of 60 mm/s.

TPU/flexible filament: Very slow print speeds (20 mm/s) are recommended due to the risk of clogs and jams.

Nylon: Print at 40 mm/s for good results, with an upper limit of 50 mm/s.

Other key element that affect printing speed are:

- Print Material and Nozzle Size
- Temperature and Cooling
- What Is a Good 3D Printing Speed
- Balancing Speed with Quality (Anker Make 2024)

Cornering Speed (Jerk/ Junction deviation)

- low acceleration and jerk ($\sim 500 \text{ mm/s}^2$; 5 mm/s) on outer walls to minimize artifacts like ghosting.
- medium acceleration and jerk ($\sim 1000\text{-}2000 \text{ mm/s}^2$; 10-15 mm/s) on inner walls and infill to optimize printing time.
- high acceleration and jerk ($\sim 3000+ \text{ mm/s}^2$; 30+ mm/s) on travel moves to reduce stringing (Here is why you should disable Junction Deviation in Marlin: [r/3Dprinting 2024](#)).

Print Speed and Acceleration

Print Speed refers to the rate at which the printer's extrusion system moves while depositing material. Measured in millimetres per second (mm/s), common print speeds range from 40 mm/s to 150 mm/s, though higher speeds may compromise quality due to reduced precision. Lower speeds generally only result in longer print times unless they drop below 5–10 mm/s, where extrusion inconsistencies can occur due to stepper motor limitations.

Acceleration defines how quickly the print head can reach its set speed. Based on the Marlin firmware standard, the default acceleration is typically 3000 mm/s^2 for rigid setups. However, many manufacturers limit this to 1000 mm/s^2 or lower to reduce vibrations and improve print quality. For our simulation, we have selected a moderate acceleration of 1500 mm/s^2 , balancing speed and stability.

At this acceleration, the time required to reach 60 mm/s from rest can be calculated using:

$$t = \frac{vf-vi}{a} = 0.04 \text{ seconds.}$$

The distance travelled during this acceleration is:

$$s = ut + \frac{1}{2}at^2 = 1.2\text{mm}$$

Junction Deviation (Jerk) Settings

Junction deviation or "jerk" refers to the minimum speed change that doesn't require acceleration. In our model, a jerk threshold of 10 mm/s is used. When changing direction, the printer decelerates to 10 mm/s before accelerating again to the target velocity of 60 mm/s.

Using our defined acceleration (1500 mm/s²), the time required to decelerate from 60 mm/s to 10 mm/s (and accelerate again) is:

$$t = \frac{60 - 10}{1500} = 0.033 \text{ seconds}$$

The distance travelled during deceleration and acceleration:

$$s = ut + \frac{1}{2}at^2 = 0.82\text{mm}$$

So, the total distance covered before and after the cornering event is approximately **1.64 mm**.

Other Factors Influencing Print Speed

- Flow Rate: Determines filament deposition and directly limits achievable print speeds.
- Layer Height & Resolution: Finer layers reduce speed but improve detail.
- Nozzle Size & Material: Larger nozzles allow faster prints; softer materials often require slower speeds.
- Temperature & Cooling: Inadequate cooling at high speeds can deform parts.

Simulation Settings Summary

TABLE 7:SIMULATION SUMMARY

Parameter	Value
Constant Print Speed	60 mm/s
Acceleration	1500 mm/s ²
Jerk Threshold	10 mm/s
Acceleration Time	0.04 seconds
Distance to Full Speed	1.2 mm
Deceleration Distance	~0.82 mm
Acceleration Distance	~0.82 mm
Total Distance (Cornering)	~1.64 mm

Definition of the Staircase Print Model

The work with Motion View initially began using primitive graphics during the early stages of Phase Two. Once all CAD parts were developed, optimized, and finalized, the CAD models were introduced as input files for simulation in Motion View. During this process, certain challenges were encountered while importing and handling the .step files. To address this issue, a decision was made to convert the .prt files into Parasolid format, which were then successfully utilized as input graphics for the simulation.

Body Groups

Initially, it was challenging to manage the multiple bodies and graphics in Motion View, as the model consisted of several detailed CAD parts. To simplify the process and make the simulation easier to handle, the components were grouped into rigid bodies.

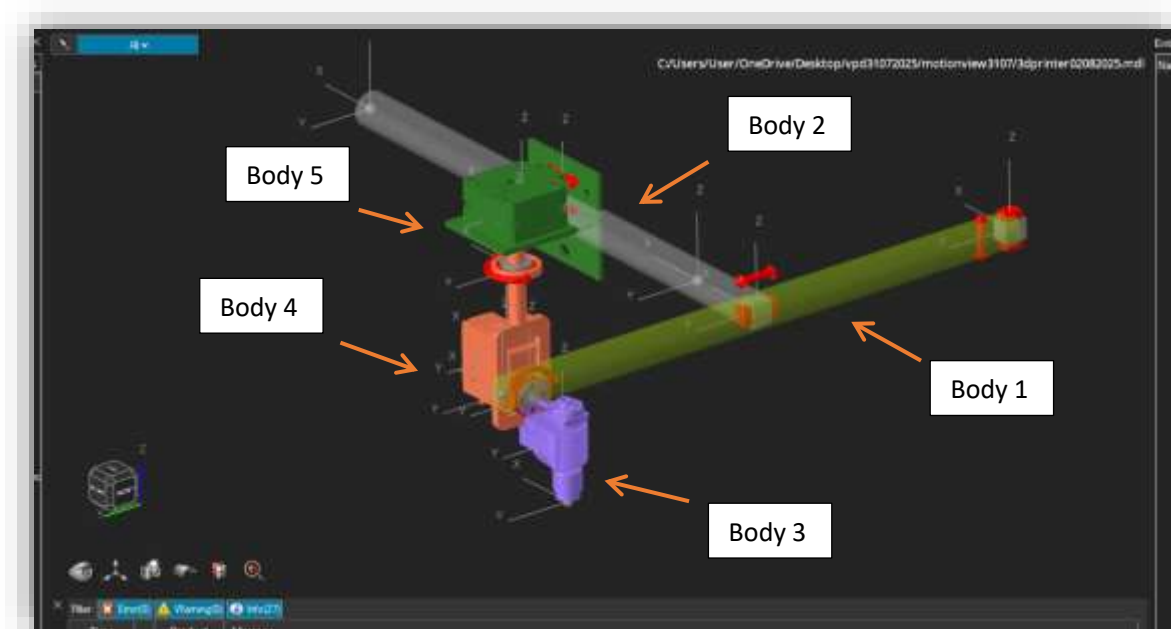


FIGURE 75: NO OF BODIES

Based on the design, five rigid groups were created as follows:

- **Body 1:** Y-rail
- **Body 2:** X-rail
- **Body 3:** Hot end and hot end holder
- **Body 4:** Servo motor and main frame
- **Body 5:** Support plate and NEMA motor

This grouping ensured that the model was simplified, well-structured, and more efficient to work with during the Motion View simulations.

Material Assignment and Mass Properties in Motion View

Upon importing the CAD model from Siemens NX into Motion View as a graphical representation, the mass and moment of inertia of the components are automatically populated. A new material, Aluminium 6061, was created in Motion View with properties such as density and modulus of elasticity, and assigned to the finger printer component. Other components, including bolts, screws, and the hot end, were assigned steel, while the nozzle was assigned brass.

Joints

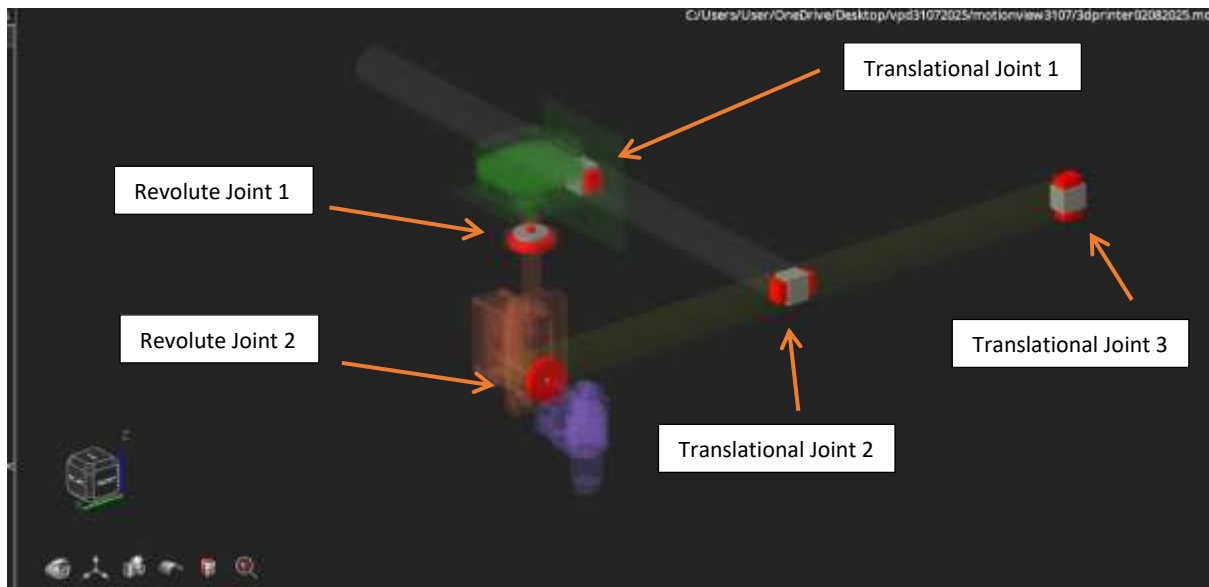


FIGURE 76:NO OF JOINTS

The finger printer CAD model is assembled in Motion View using the following joints:

1. Revolute Joint 1: Provides rotation of the finger printer along the global Z-axis.
2. Revolute Joint 2: Allows rotational motion of the hot end mount relative to the main frame of the finger printer along the global Y-axis.
3. Translational Joint 1: Enables translational motion of the finger printer along the global X-axis.
4. Translational Joint 2: Enables translational motion of the finger printer along the global Y-axis.
5. Translational Joint 3: Provides translational motion of the bed along the global Z-axis.

TABLE 8:SIMULATION SUMMARY

Type of joint	Degree of freedom	Constraints degrees of freedom
Revolute Joint 1	1	5
Revolute Joint 2	1	5
Translational Joint 1	1	5
Translational Joint 2	1	5
Translational Joint 3	1	5

Total no. of bodies = 5

1. Y-rail
2. X-rail
3. Hot end and hot end holder
4. Servo motor and main frame
5. Support plate and NEMA motor

Total Degrees of Freedom from 5 bodies = 30

Degrees of Freedom constrained by the joints = 25

Resultant degrees of freedom of the nozzle = 5 (Translational along X, Y, Z and Rotational about Y and Z axes.

Trajectory of the staircase printing

In this project, the staircase is not built through stiff, step-by-step movements, but through a smooth, flowing curve-based trajectory. Instead of breaking the motion into rigid segments, we guided the nozzle with a continuous path — almost like sketching the staircase by hand. The motion data, defined in terms of positions (x, y, z) and orientation (rx), allows the printer to move naturally as it builds each layer.

The nozzle sweeps across a horizontal range of about 50–190 mm in X and 0 to –80 mm in Y, covering the base outline of the staircase. At the same time, the Z-axis motion climbs gradually from 0 to 40 mm, stacking the layers one by one. In our setup, this vertical motion isn't provided by a moving bed. Instead, it comes from the translational joint between the Y-rail and the ground body, which lowers the assembly slightly after each layer is complete.

What makes the motion particularly striking is the way the nozzle changes its angle as the staircase grows. It begins upright at 0° , perfect for the vertical walls, then gently tilts to about 55° for the inclined faces, and finally lies flat at 90° to smooth across the top surface. Think of it like a hand holding a pen: straight up for drawing lines, then gradually angling down for shading.

The entire path unfolds over about 142 seconds, with the nozzle gliding at a steady pace. The print begins by tracing the curved outline of the base in the X–Y plane, then rises layer by layer, adjusting its tilt as it goes. This continuous, curve-based motion creates smoother transitions between vertical and horizontal surfaces compared to the older, step-wise approach.

All of this takes place comfortably within the $250 \times 250 \times 300 \text{ mm}^3$ workspace of the Tronxy X5SA PRO printer, showing that a curve-driven approach can bring both elegance and precision to the task of staircase printing.

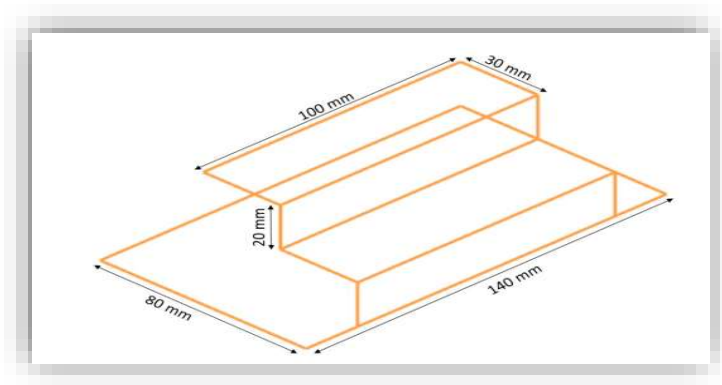


FIGURE 77: DIMENSIONS OF STAIR CASE

Staircase Simulation:

In Phase 3, the objective is to maintain a **constant velocity of the print head nozzle** during the simulation. For the inverse kinematics simulation of the Stair model, **predefined motion curves** are used.

Staircase Model Description for Simulation

The staircase model consists of two primary steps with the following dimensions:

- **First step:** 100 mm long, 30 mm wide, and 20 mm high
- **Second step:** identical dimensions, stacked above the first step
- **Base platform:** 140 mm × 80 mm

The print head is simulated to traverse these steps using the motion curves, moving along **X, Y, and Z axes** according to the step orientation. The motion ensures smooth acceleration and deceleration along vertical and horizontal surfaces, maintaining near-constant velocity as the nozzle moves from the base to the top step.

TABLE 9:STAIR CASE TABLE

S.NO	Motion	Expression
1	X MOTION TRANSLATION	Input using curve
2	Y MOTION TRANSLATION	Input using curve
3	Z MOTION TRANSLATION	Input using curve
4	HOTEND ROTATION	Input using curve
5	MAIN FRAME ROTATION	Input using curve

Proof of Print:

Proof of print operability in MotionView essentially means demonstrating that the sample can be printed smoothly in the simulation without any collisions or interruptions, as illustrated in the figures below.

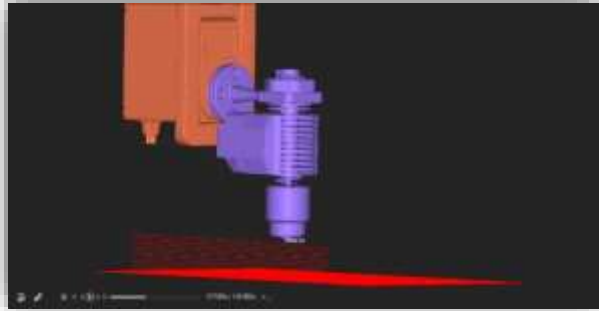


FIGURE 78: PROOF OF PRINT VIEW1.

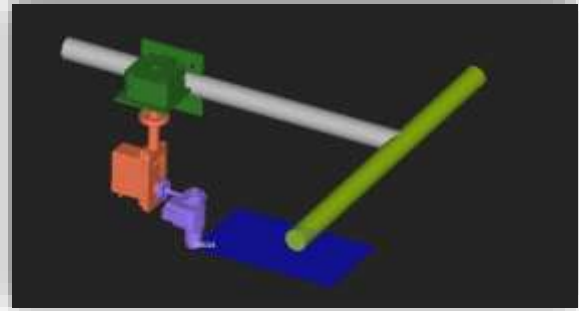


FIGURE 79: PROOF OF PRINT VIEW2

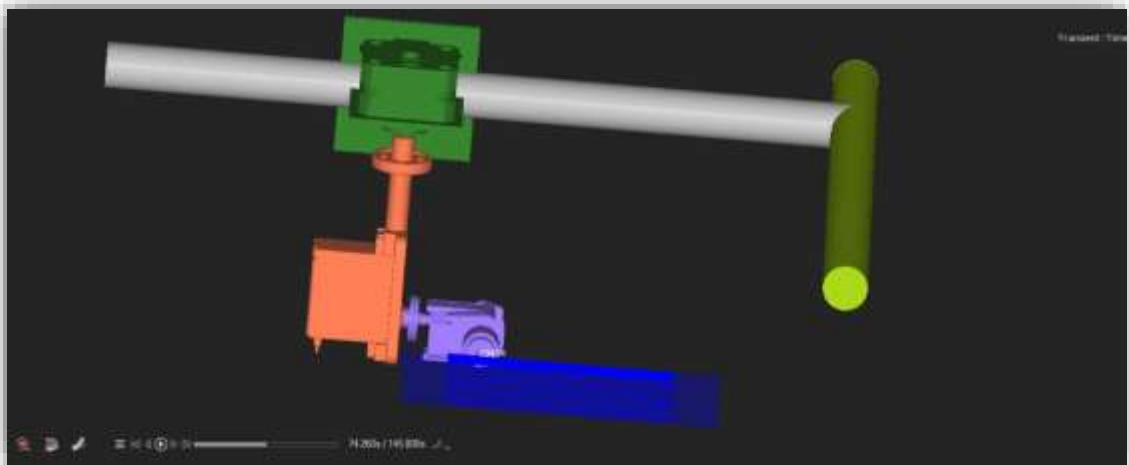


FIGURE 80: PROOF OF PRINT VIEW3

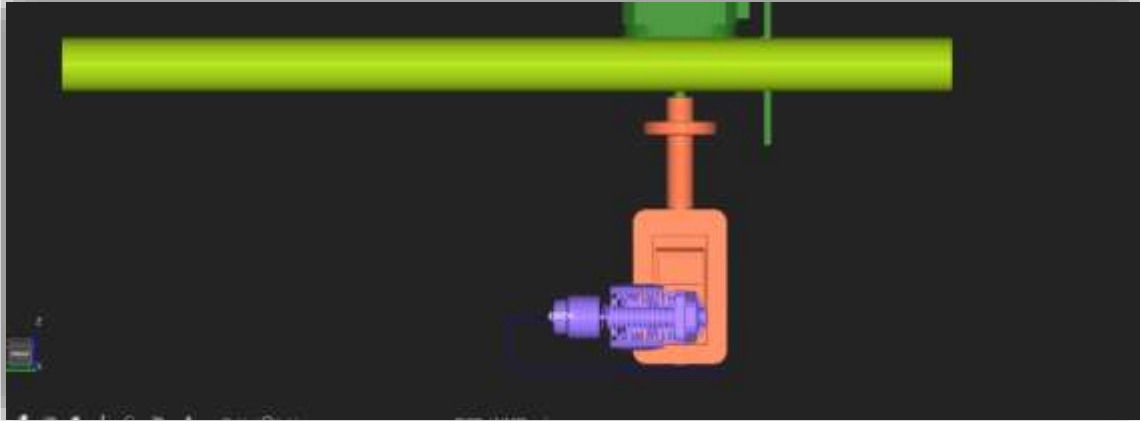


FIGURE 81: PROOF OF PRINT VIEW4

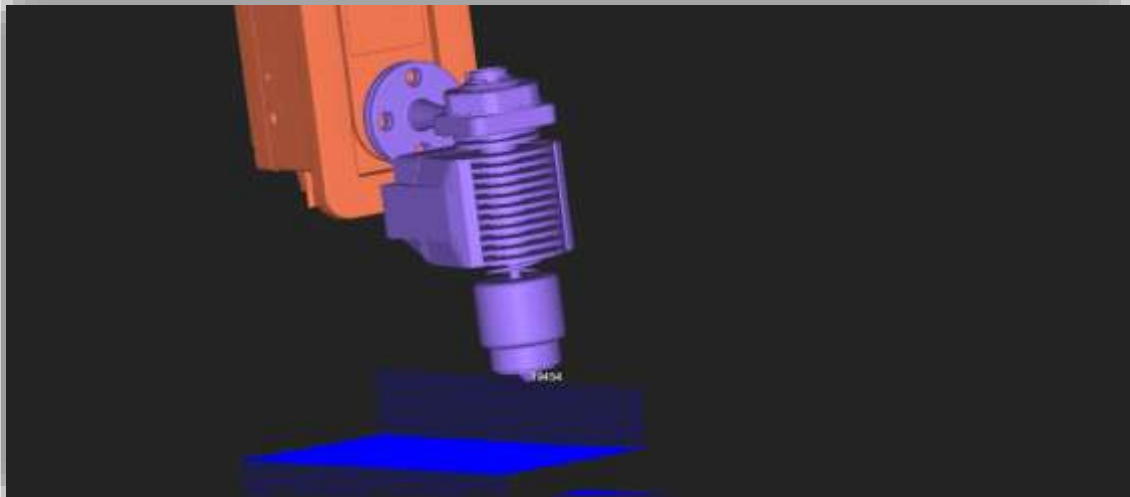


FIGURE 82: PROOF OF PRINT VIEW5

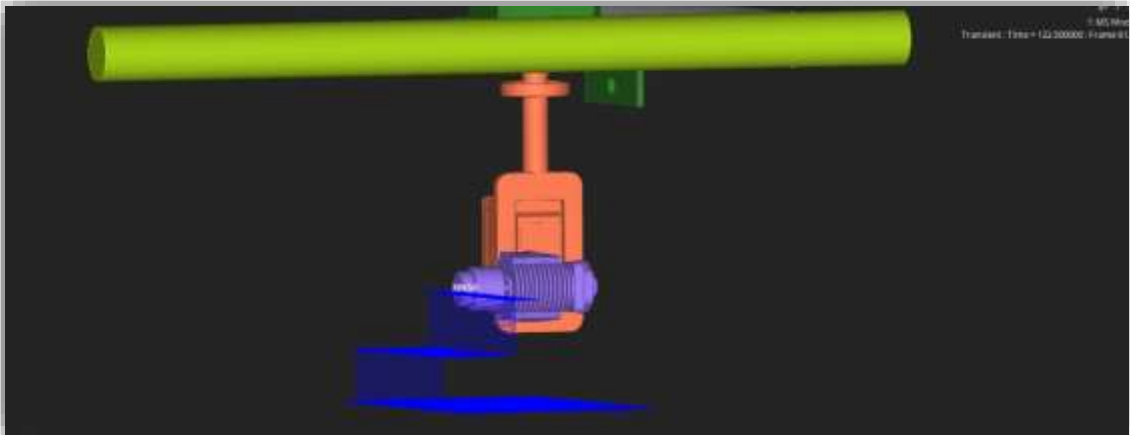


FIGURE 83: PROOF OF PRINT VIEW6

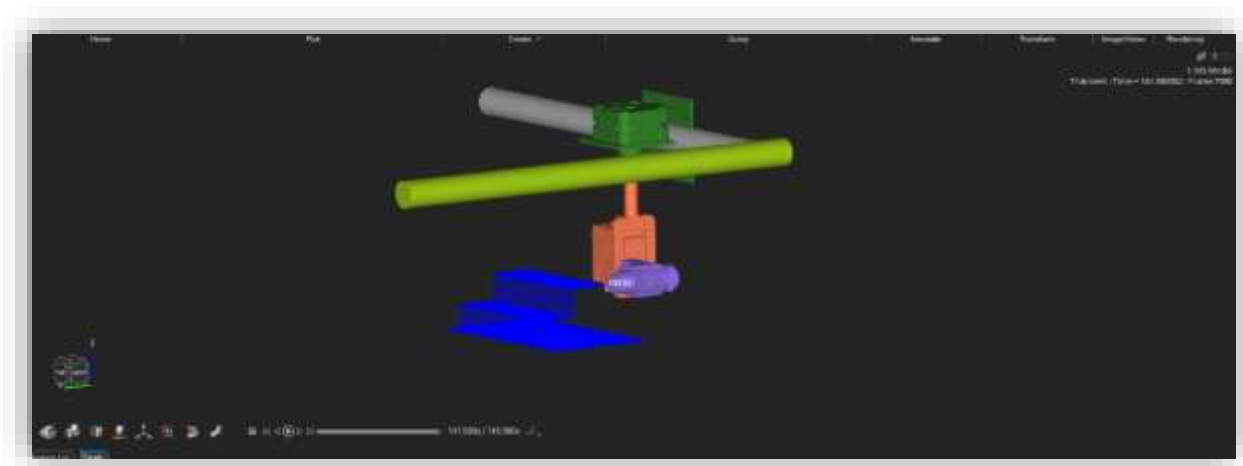


FIGURE 84: PROOF OF PRINT VIEW7

Joint Space Analysis:

To ensure that the end-effector follows the desired path, it is crucial to control each joint with precision. This level of control is achieved through **joint space analysis**. For this purpose, the motion of the joints is driven by the **output curves generated from the staircase inverse kinematics simulation**.

TABLE 10:ABOUT JOINTS.

S.NO	Motion	Direction	Expression
1	X MOTION TRANSLATION	Translation in X	Input using curve
2	Y MOTION TRANSLATION	Translation in Y	Input using curve
3	Z MOTION TRANSLATION	Translation in Z	Input using curve
4	HOTEND ROTATION	Rotation about X	Input using curve
5	MAIN FRAME ROTATION	Rotation about Z	Input using curve

Curves used for Joint Space analysis:

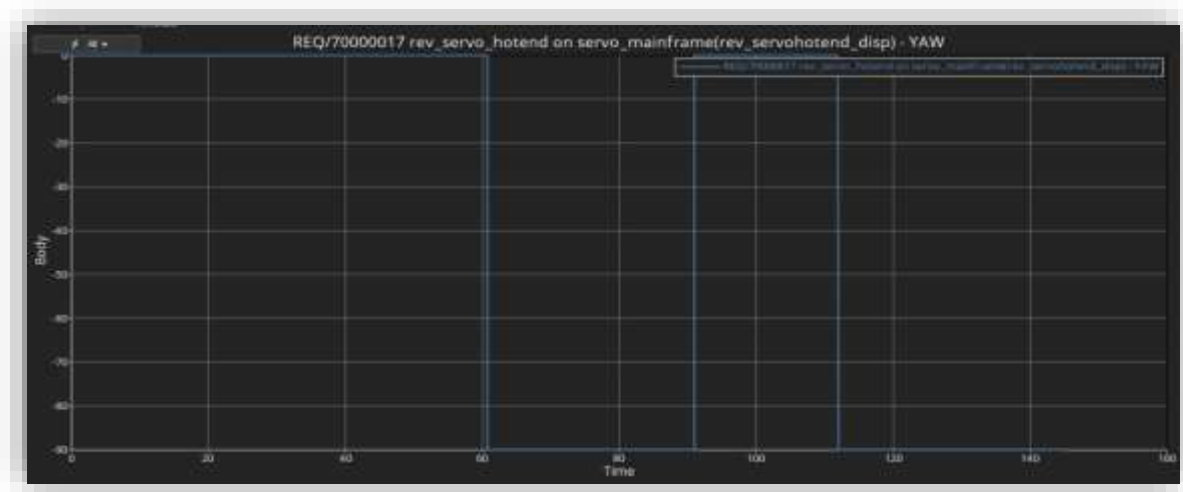


FIGURE 85: ANGULAR DISPLACEMENT IN X VS TIME

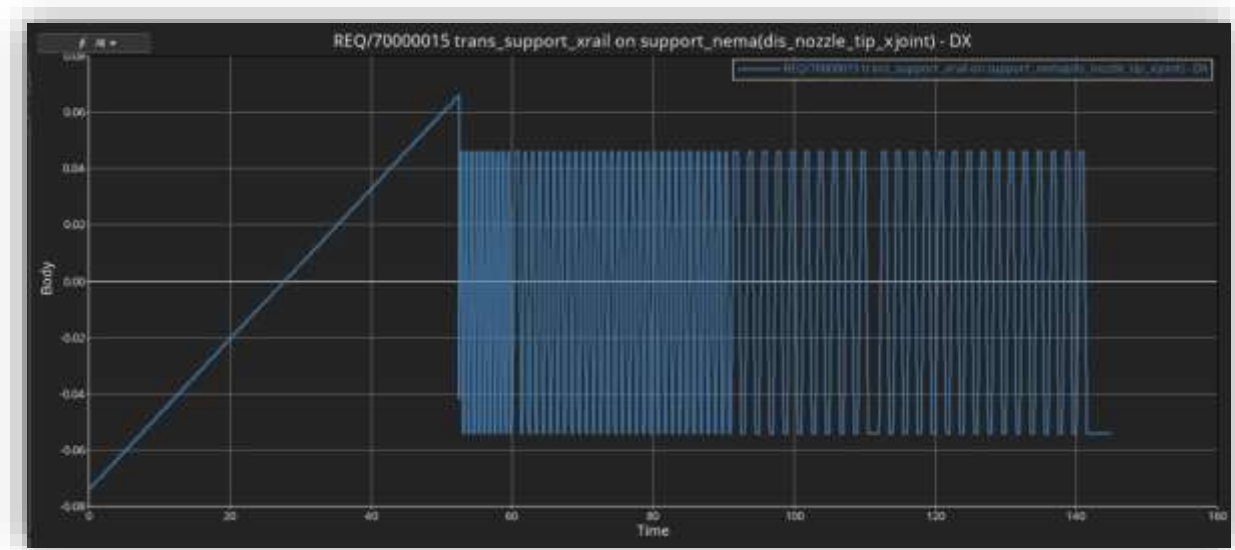


FIGURE 86: TRANSLATION IN X

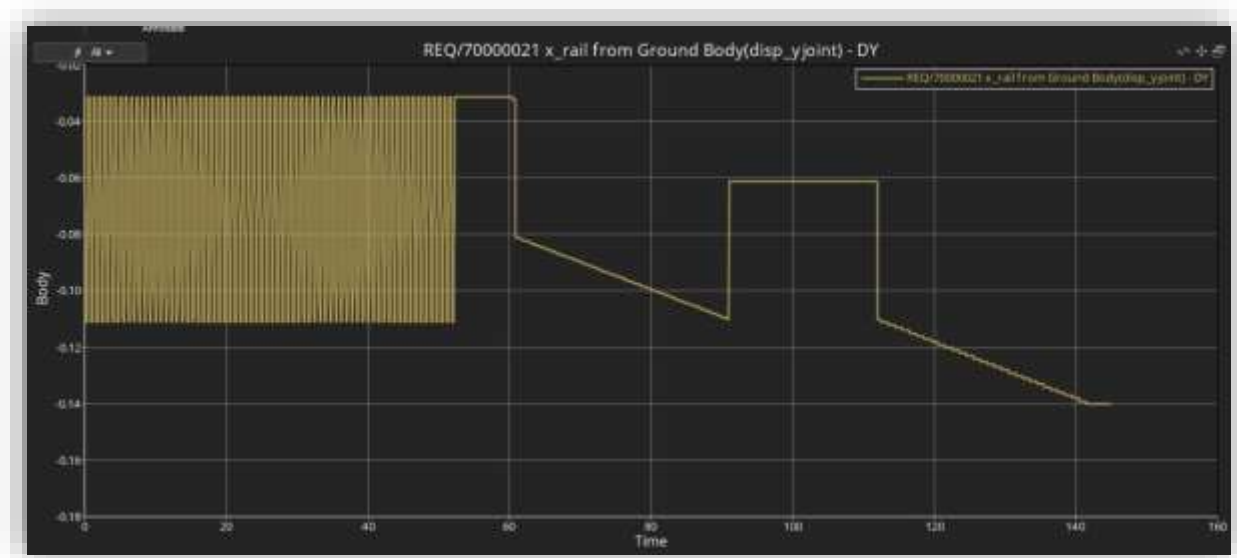


FIGURE 87: TRANSLATION IN Y

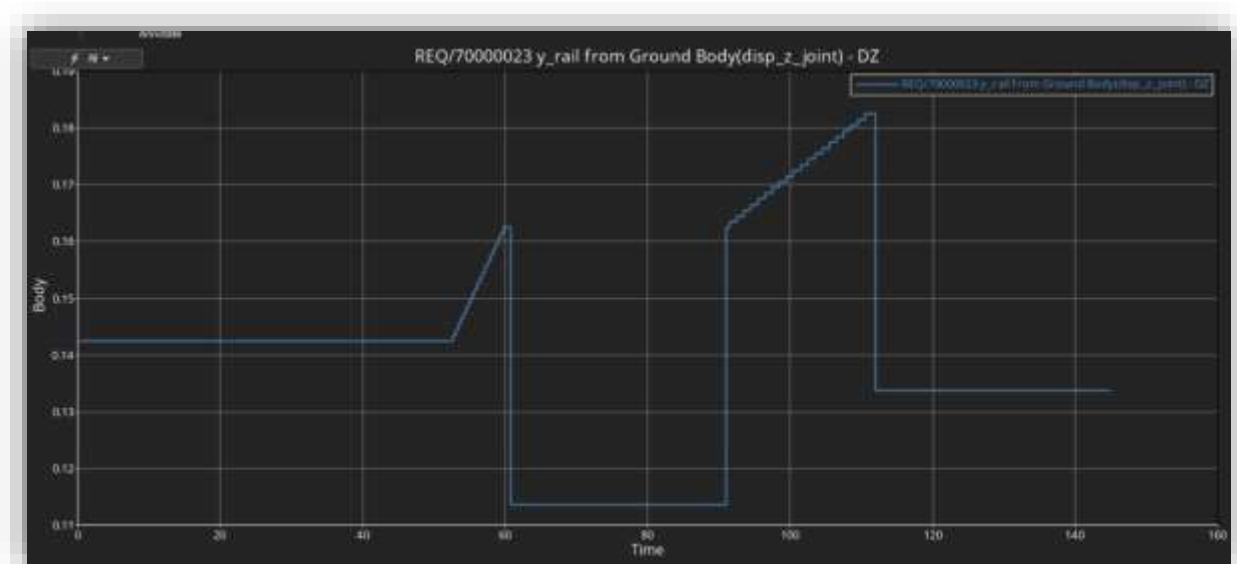


FIGURE 88: TRANSLATION IN Z

Proof of Drive Dimensioning: -

In MotionView, proof of drive dimensioning refers to verifying that the selected drive systems (motors or actuators) are properly sized and have sufficient capacity to withstand the torque and forces generated during operation. This step ensures that the drives meet the performance requirements for the intended application.

In our model, the **upper motor** is a NEMA 17 stepper motor with a maximum holding torque of **0.15 Nm**. The torque required for our application is only **0.029 Nm**, which is well within the motor's rated capacity, providing a safe operating margin. The **lower actuator** is a **Servo**

DS3218 PRO, which has a maximum torque capacity of approximately **2.3 Nm**. while the system requires only **0.019 Nm**.

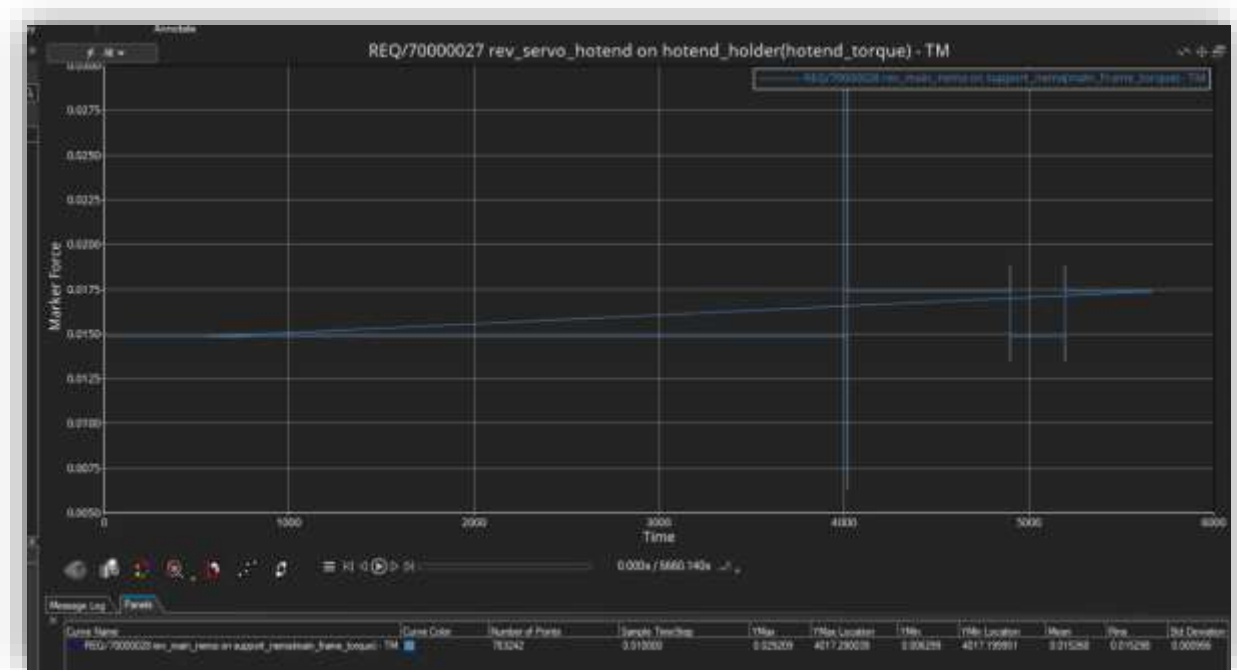


FIGURE 89: TORQUE ACTING ON TOP STEPPER MOTOR

TABLE 11:MAX AND MIN TORQUE ON UPPER STEPPER

Maximum Torque on Upper Stepper motor	Minimum Torque on Upper Stepper motor	Capacity of Upper Stepper Motor
0.029 Nm	0.006299 Nm	0.15 Nm

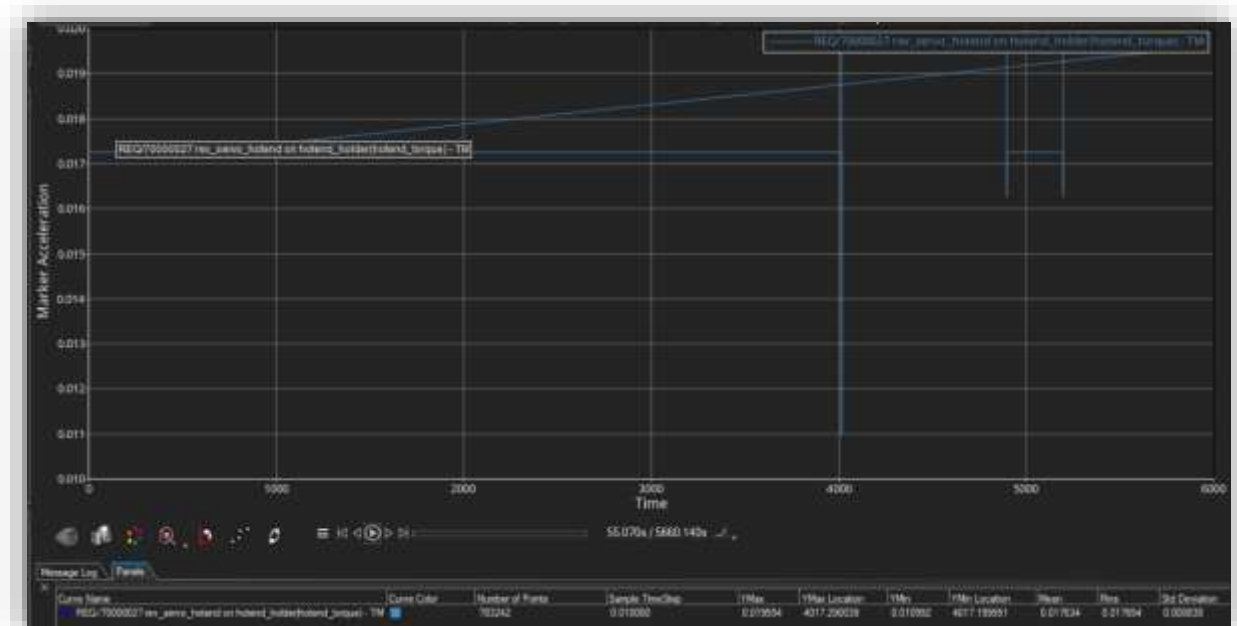


FIGURE 90: TORQUE ACTING ON LOWER SERVO MOTOR.

TABLE 12:MAX AND MIN TORQUE ON LOWER SERVO

Maximum Torque on Lower Servo motor	Minimum Torque on Lower Servo motor	Capacity of Lower Servo Motor
0.019 Nm	0.010992Nm	2.3 Nm

Free Form Geometry with Overhang:

Trajectory of the Flower vase:

The figure below shows the vase toolpath generated from our code (spiral trajectory).

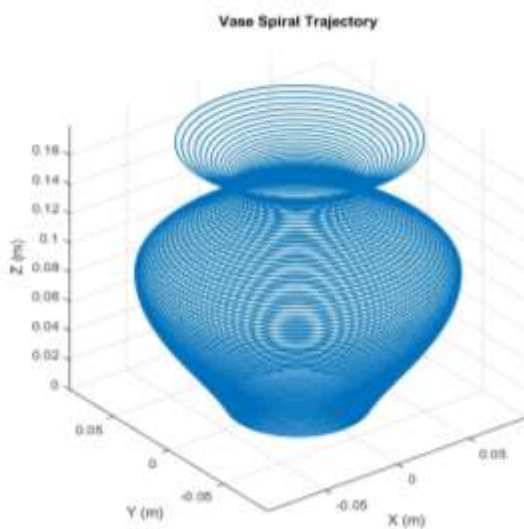


FIGURE 91:VASE SPIRAL TRAJECTORY

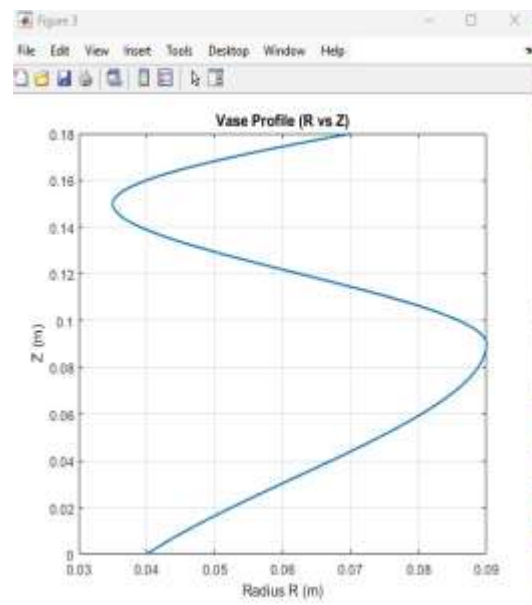


FIGURE 92: VASE PROFILE CURVE

Path Creation Using MATLAB:

The 3D path for the printer's end-effector was generated using a MATLAB script. This script provided flexibility to define the vase contour, set the layer height, and control the orientation of the nozzle. By combining radius functions with spiral motion, the toolpath produces a slim vase with bulges.

Parameter Definitions:

The key parameters in the MATLAB and MotionView expressions that define the vase geometry are:

Base Radius ($R_{base} = 45 \text{ mm}$): Sets the starting diameter at the bottom of the vase.

Bulge Amplitudes ($a_1 = 12$ mm, $a_2 = -9.9$ mm, $a_3 = 8.4$ mm): Define how the radius expands or contracts at different heights. Positive values create outward bulges, while negative values create a neck (constriction).

Gaussian Centers: Lower bulge centered at 12.5% of height.

Neck centered at 20% of height.

Upper bulge centered at 25% of height.

Print Height ($H = 250$ mm): Vertical distance from base to lip.

Time ($T = 90$ s): Total printing time.

Angular Speed ($\omega = 43.63$ rad/s): Rotation rate that defines how fast the nozzle spins around the vase.

X Offset (165 mm): Moves the vase away from the machine origin for printing.

Vase Body Path Generation:

Radius Function (R): The vase's contour is defined by a base radius plus

Gaussian terms:

$$R(t) = base + a_1 e^{-(ct-0.125)^2 / 2\sigma^2} - a_2 e^{-(ct-0.200)^2 / 2\sigma^2} + a_3 e^{-(ct-0.250)^2 / 2\sigma^2}$$

$$R(t) = R_{base} + a_1 e^{-\frac{(ct-0.125)^2}{2\sigma^2}} - a_2 e^{-\frac{(ct-0.200)^2}{2\sigma^2}} + a_3 e^{-\frac{(ct-0.250)^2}{2\sigma^2}}$$

where $c = 0.002777778$ $c = 0.002777778$ scales time into normalized height. This creates the lower bulge, slim neck, and upper bulge.

$$\text{Helical Spiral: } x(t) = 0.165 + R(t) \cos\left[\frac{2\pi}{T} t\right], y(t) = R(t) \sin\left[\frac{2\pi}{T} t\right], z(t) = 0.002777778t$$

$$t \quad x(t) = 0.165 + R(t) \cos(\omega t), y(t) = R(t) \sin(\omega t), z(t) = 0.002777778t$$

The nozzle spirals upward while adjusting its radius, forming the vase wall. Layer-by-Layer Motion: The spiral replaces discrete layer-by-layer circles. Each revolution adds a small increment in Z , determined by the vertical rise rate (≈ 0.00278 m/s).

Neck and Lip Formation:

Neck ($Z \approx 0.20H$):

The negative Gaussian reduces radius to ~ 35 mm, forming a narrow waist.

Lip Flare ($Z \approx H$):

The upper Gaussian restores radius to ~ 50 mm at the mouth, giving a natural vase opening.

Smoothness:

Gaussian functions ensure transitions are continuous, preventing sharp corners or discontinuities in the path.

End-Effector Orientation:

Rotation about X (Rx):

Held constant at 45° (0.785 rad) to demonstrate the nozzle's ability to tilt and improve layer adhesion.

Rotation about Z (Rz):

Defined as:

$$Rz(t) = \omega t + \pi/2$$

This keeps the nozzle tangent to the spiral toolpath at all times, aligning deposition direction with the trajectory.

Overhang Handling:

By combining Rx tilt and the radius expansion (bulges), the nozzle maintains proper deposition even as the vase overhangs outward beyond the neck region.

Key Differences from Hemisphere–Cylinder–Disc Case:

Sphere/Cylinder/Disc (seniors): Piecewise geometry with sharp transitions at boundaries.

Vase (our case): Continuous spiral with smooth Gaussian-defined bulges (no sudden radius jumps).

Orientation:

Seniors used different orientations for hemisphere, cylinder, and disc.

We use continuous Rz tangent following and a fixed Rx tilt for smoother control.

Proof of Print:

Proof of print operability in MotionView essentially means demonstrating that the sample can be printed smoothly in the simulation without any collisions or interruptions, as illustrated in the figures below.

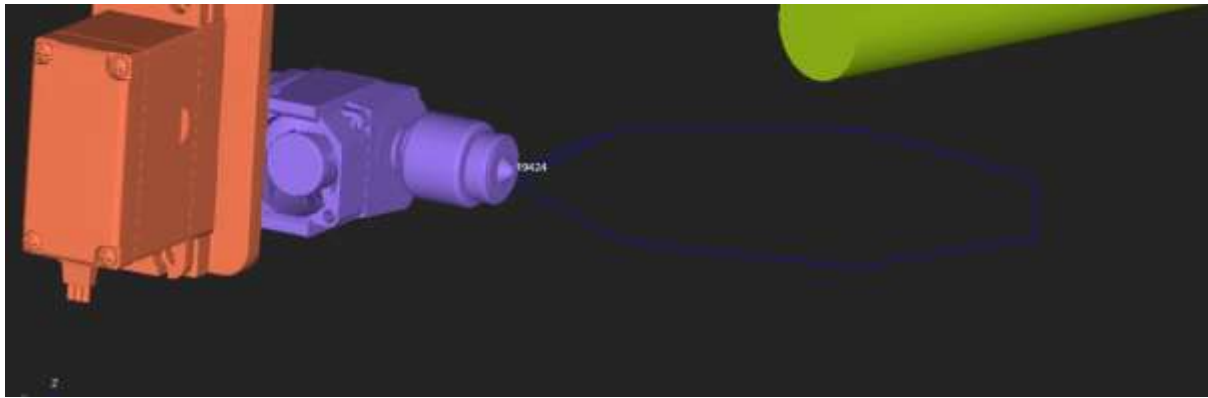


FIGURE 93:PROOF OF PRINT VIEW 1

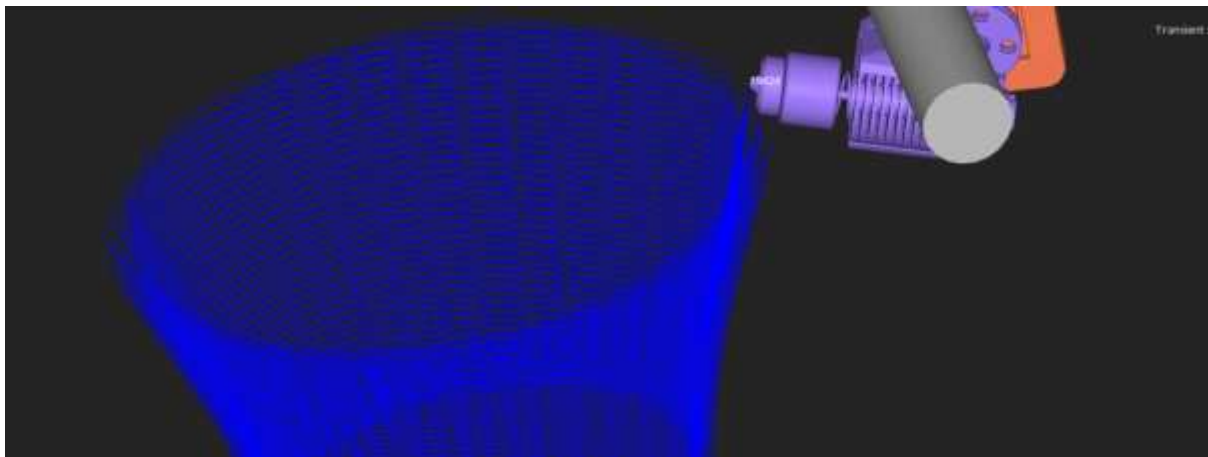


FIGURE 94:PROOF OF PRINT VIEW 2

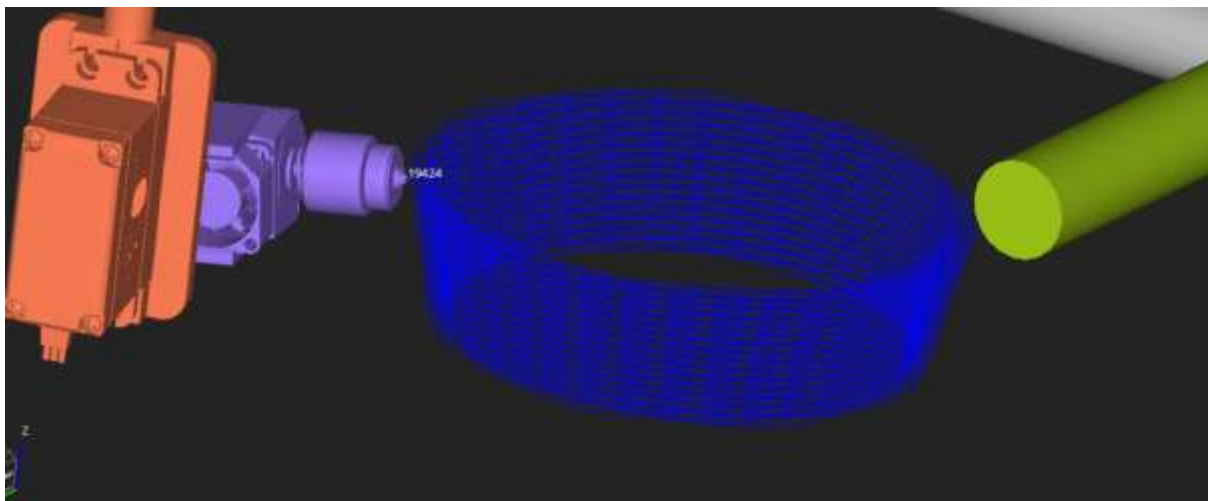


Figure 95: Proof of Print Vie

w 3

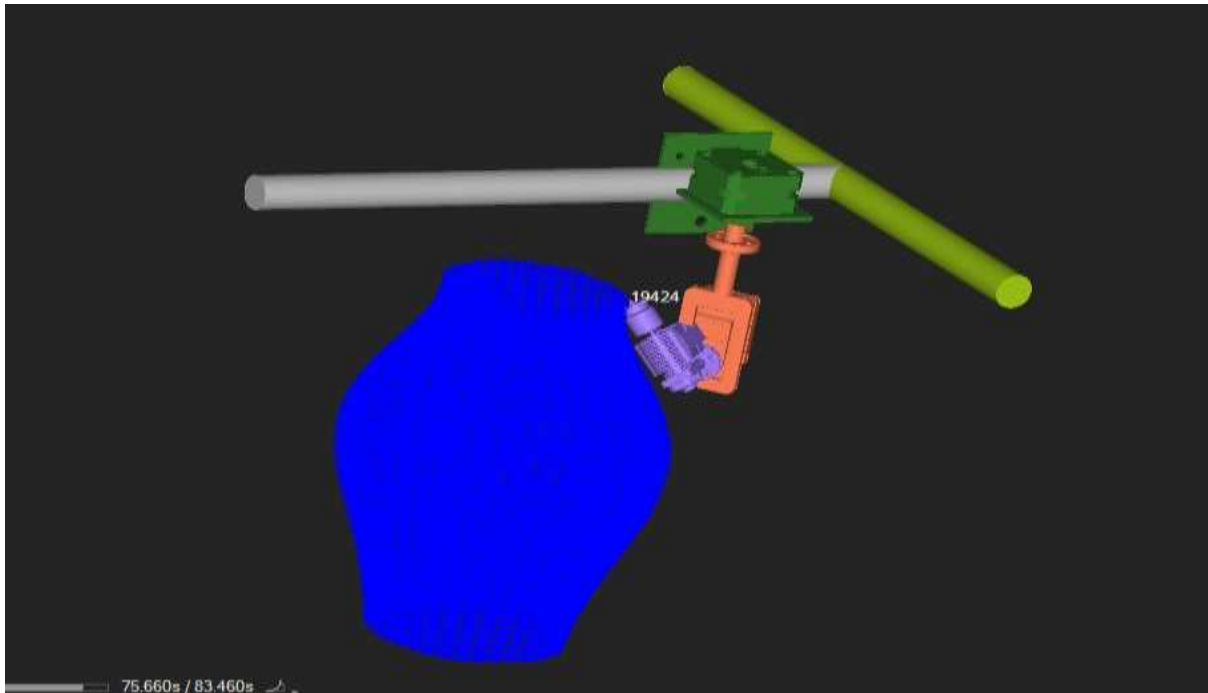


FIGURE 96:PROOF OF PRINT VIEW 4

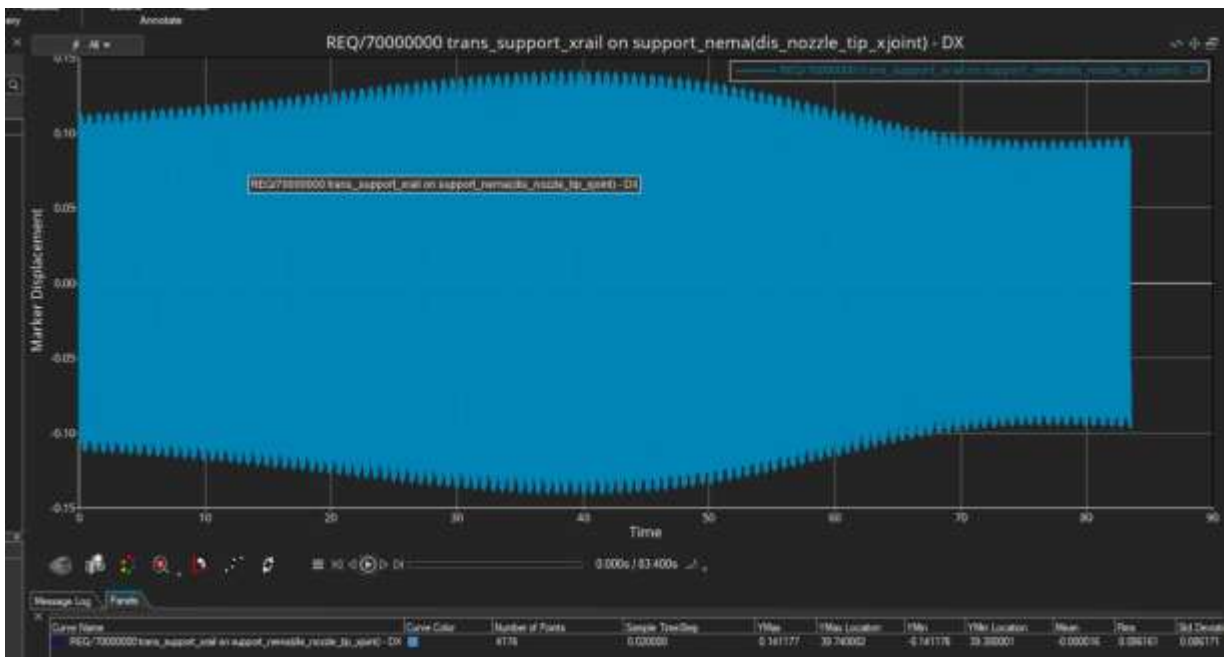


FIGURE 97:DISPLACEMENT DX

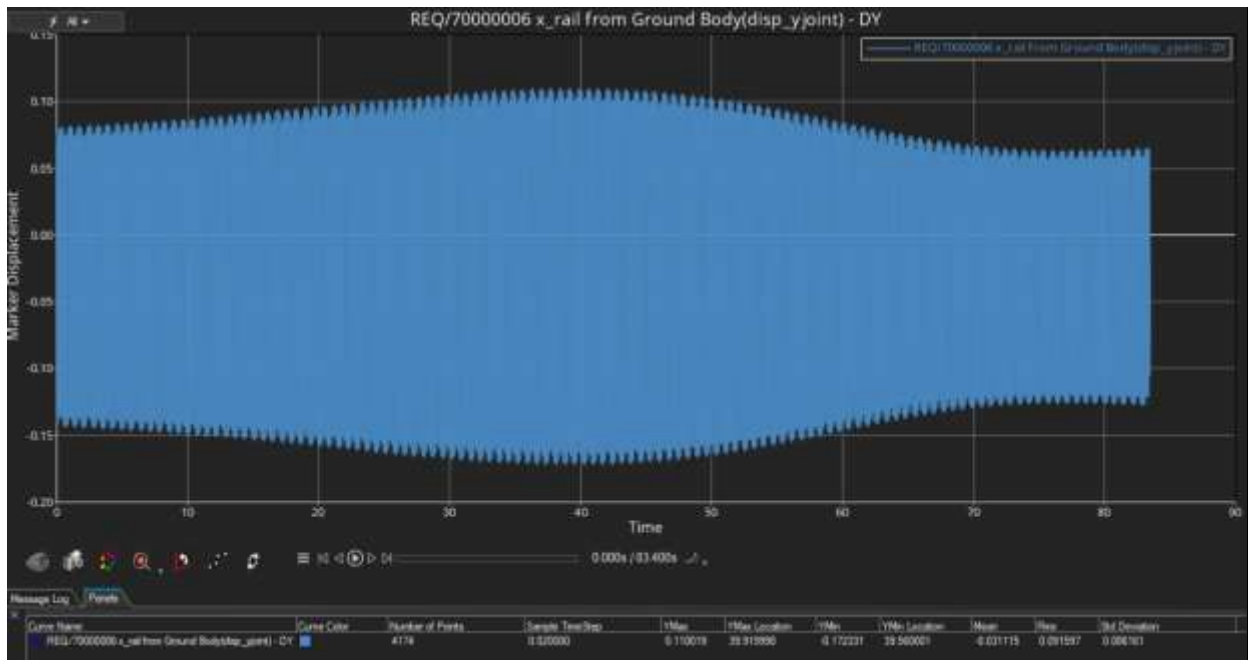


FIGURE 98:DISPLACEMENT DY

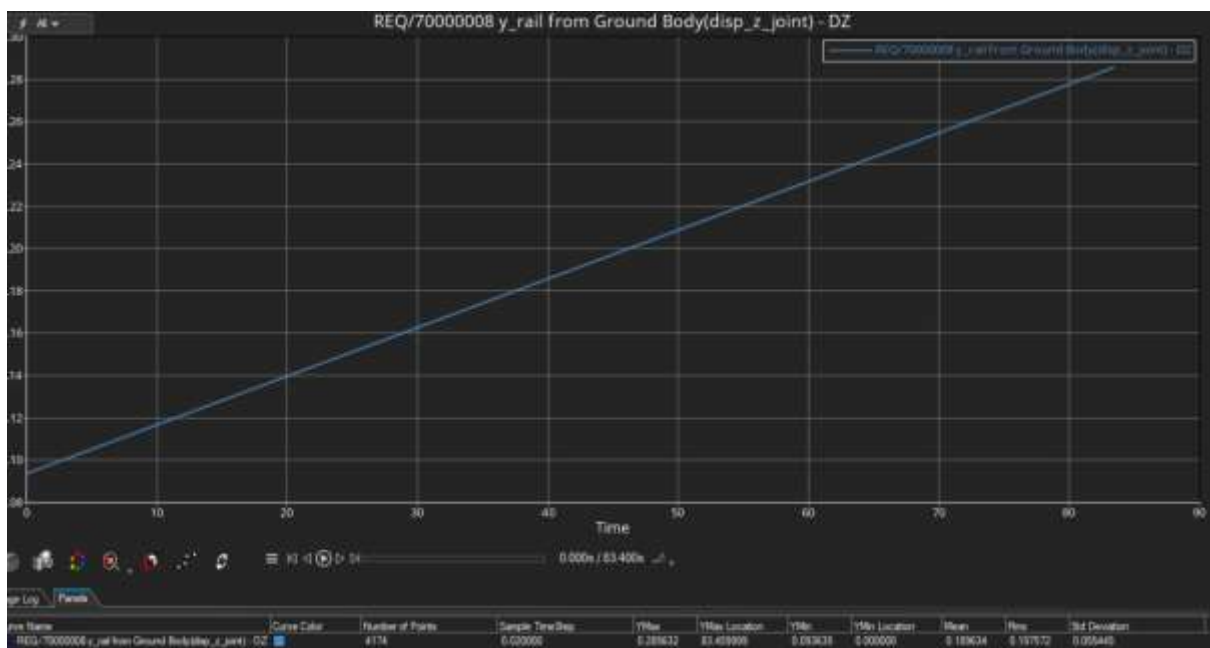


FIGURE 99:DISPLACEMENT Dz

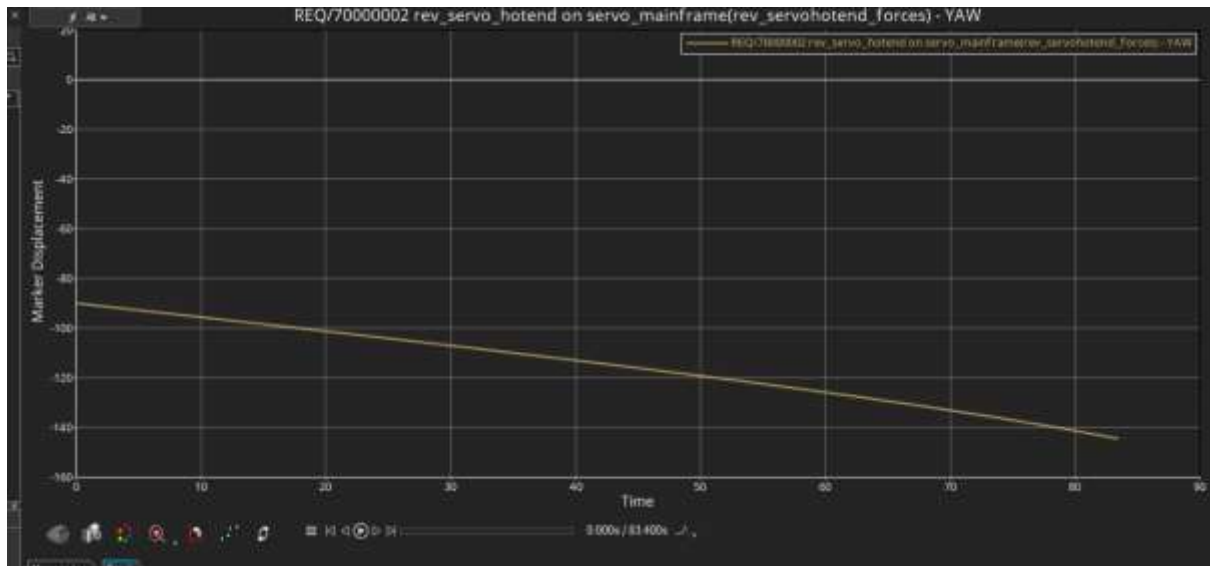


FIGURE 100: ANGULAR DISPLACEMENT

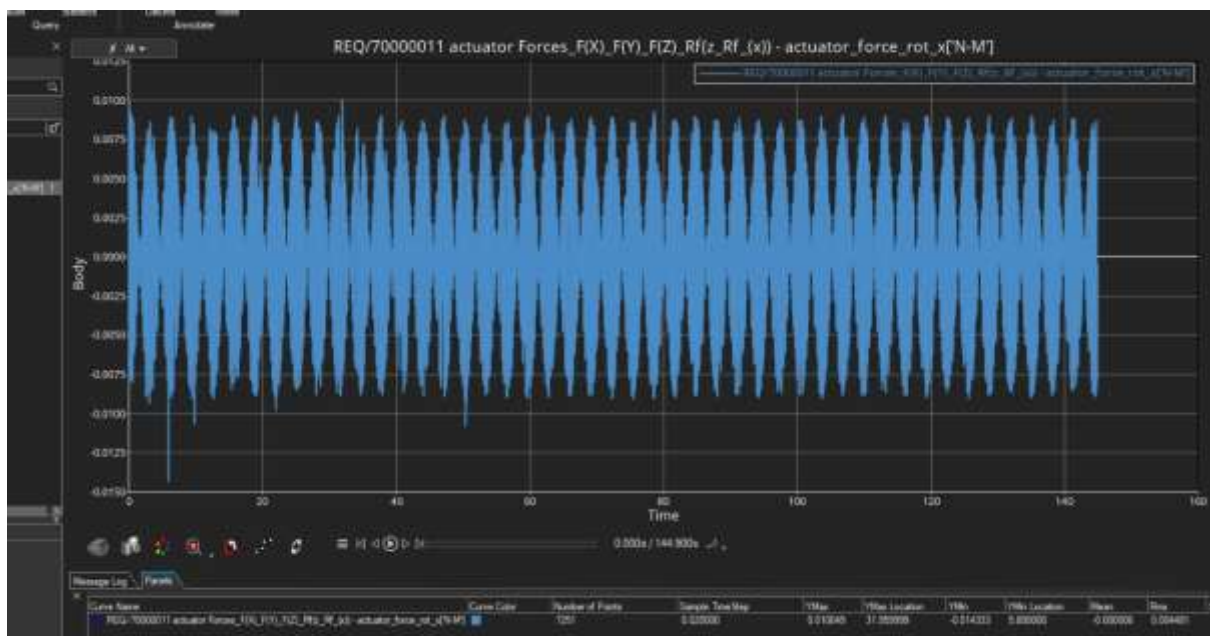
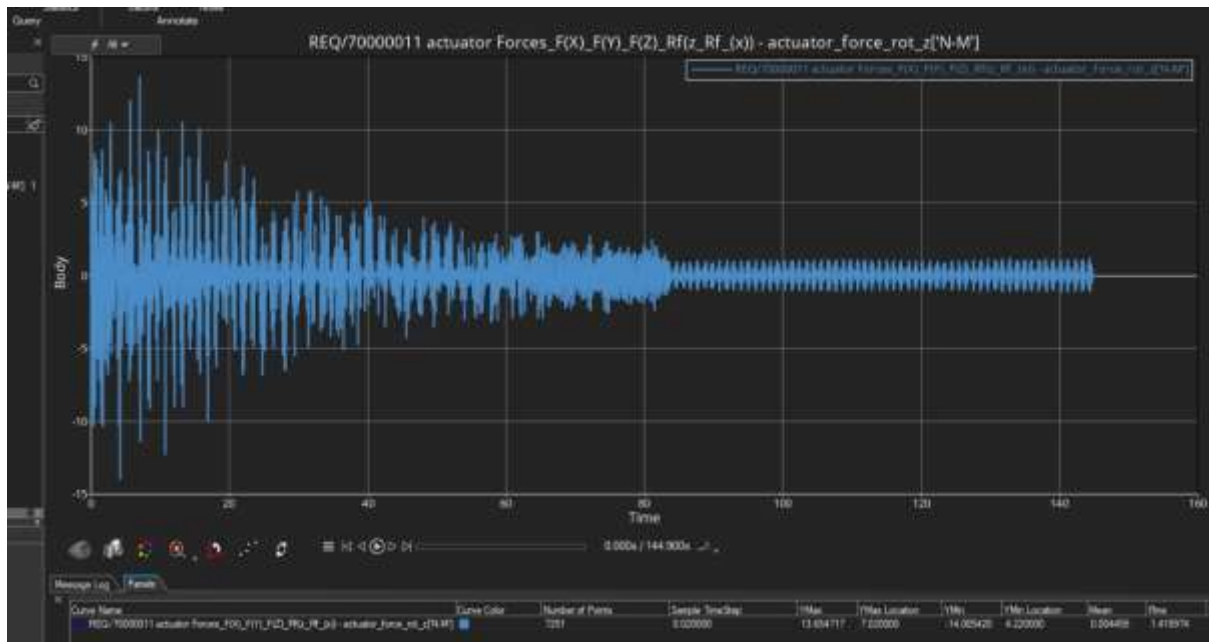


FIGURE 101:HOTEND TORQUE

TABLE 13:MAXIMUM AND MINIMUM TORQUES ON LOWER MOTOR

Maximum Torque on Lower Stepper motor	Minimum Torque on Lower Stepper motor	Capacity of Lower Stepper Motor
0.019 Nm	0.010049 Nm	-0.01433 Nm



Maximum Torque on Upper Stepper motor	Minimum Torque on Upper Stepper motor	Capacity of Upper Stepper Motor
0.029 Nm	7.02 Nm	-14.005 Nm

Make-or-Buy Analysis: Cost Estimation

1.Support Plate

- Weight – 0.0283 Kg.
- Material Required – 0.411 Kg and waste material out of it will be 0.3827 Kg.
- Raw Material Cost – Aluminium 6061 5.83 Euro/Kg, so $0.411 \times 5.83 = 2.40$ Euro
- Scrap Value – 1.50 Euro/Kg, Value of removed material $0.3827 \times 1.5 = 0.57$ Euro
- Machining and Labour – 22 Euros overall
- Final – $2.40 + 22 - 0.57 = \mathbf{23.83 \text{ Euro}}$

2.Main Frame

- Weight – 0.0179 Kg.
- Material Required – 0.230 Kg and waste material out of it will be 0.2121 Kg.
- Raw Material Cost – Aluminium 6061 5.83 Euro/Kg, so $0.230 \times 5.83 = 1.34$ Euro.
- Scrap Value – 1.50 Euro/Kg, Value of removed material $0.2121 \times 1.5 = 0.32$ Euro.
- Machining and Labour – 35 Euros overall. (CNC operator 60 Euro/hr approx. 35 min required for setting up and machining in all).
- Final – $1.34 + 35 - 0.32 = \mathbf{36.02 \text{ Euro.}}$

3.Hotend Mount

- Weight – 0.0029 Kg.
- Material Required – 0.0419 Kg and waste material out of it will be 0.039 Kg.
- Raw Material Cost – Aluminium 6061 5.83 Euro/Kg, so $0.0419 \times 5.83 = 0.24$ Euro.
- Scrap Value – 1.50 Euro/Kg, Value of removed material $0.039 \times 1.5 = 0.0585$ Euro.
- Machining and Labour –50 Euros overall.
- Final – $0.24 + 50 - 0.0585 = \mathbf{50.18 \text{ Euro}}$.

4.Rigid Flange Coupling

- Final – **8.55 Euro**. (Buy Part)

5.mg996r round servo horn

- Final – **1.509 Euro** (Buy Part)

6.Nema 17 Stepper Motor 17HS4023

- Final – **8.49 Euro** (Buy Part)

7.Servo Motor DS3218 PRO Servo Motor

- Final – **11.99 Euro** (Buy Part)

8.Hotend E3D Revo

- Final – **106.99 Euro** (Buy Part)

9. DIN 912 M3x8 - 12 Nos*0.11 Euro each = **1.32 Euro**

DIN 912 M4x8 – 4 Nos*3.61 Euro each = **14.44 Euro**

DIN 934 M3 – 8 Nos*1.43 Euro = **11.44 Euro**

- Final – **27.2 Euro** (Buy Part)

Total Cost of Printer = $23.83 + 36.02 + 50.18 + 8.55 + 1.509 + 8.49 + 11.99 + 106.99 + 27.2 = \mathbf{274.759 \text{ Euros}}$.

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Contribution Of Team Members

	Madana Gopala Reddy Konda	Mithin Kumar Chithirala	Yuvraj Ramchandra Labhade	Vinay Kumar Pepeti	Sum
Initial Concept Design	25	20	30	25	100
CAD Design	30	18	15	37	100
Documentation	16	28	35	21	100
Topology Optimization	20	20	22	38	100
MBS	43	16	21	20	100
Overall	26.8	20.4	24.6	28.2	100

Signature				
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With our signature we declare, that we agree with the statements of all participants.